NEIGH CORP



Neigh Corp - Golden Meadows

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NEIGH CORP



1 Introduction

This section is to give a general introduction to the company, including a description of its main product offerings, markets, and strategy.

Neigh Corp is a small game company aiming to create its first horse-themed game. The product is an MMORPG with "gacha mechanics", a video game system in which the player spends in-game currency to receive a random in-game item, that combines various semi-realistic, real-life horse-related activities into a single game. The game is designed to be refreshing, immersive, and modern, and allows the player to roleplay as a horse trainer, enjoy riding in an open world, take part in competitions, and gather and sell horses and horse equipment. The game will have a lively community active in the game forum.

The desired customer base consists mostly of teenagers and young adults who are interested in horses. Neigh Corp welcomes experienced gamers as well as players who are completely new to gaming.

The primary sources of income are a recurring premium membership subscription that is offered to players, as well as in-game cosmetic purchases and microtransactions for in-game advantages.

2 Value Network Analysis

This section is to show the value network in which the company will work. Agents, resources, and exchanges of resources are to be made explicit. Note that the project group here has to clarify exactly which products the company is to offer. Also note that the value network analysis has to take into account that the company is to work in different markets with different actors.

2.1 VDML Diagram

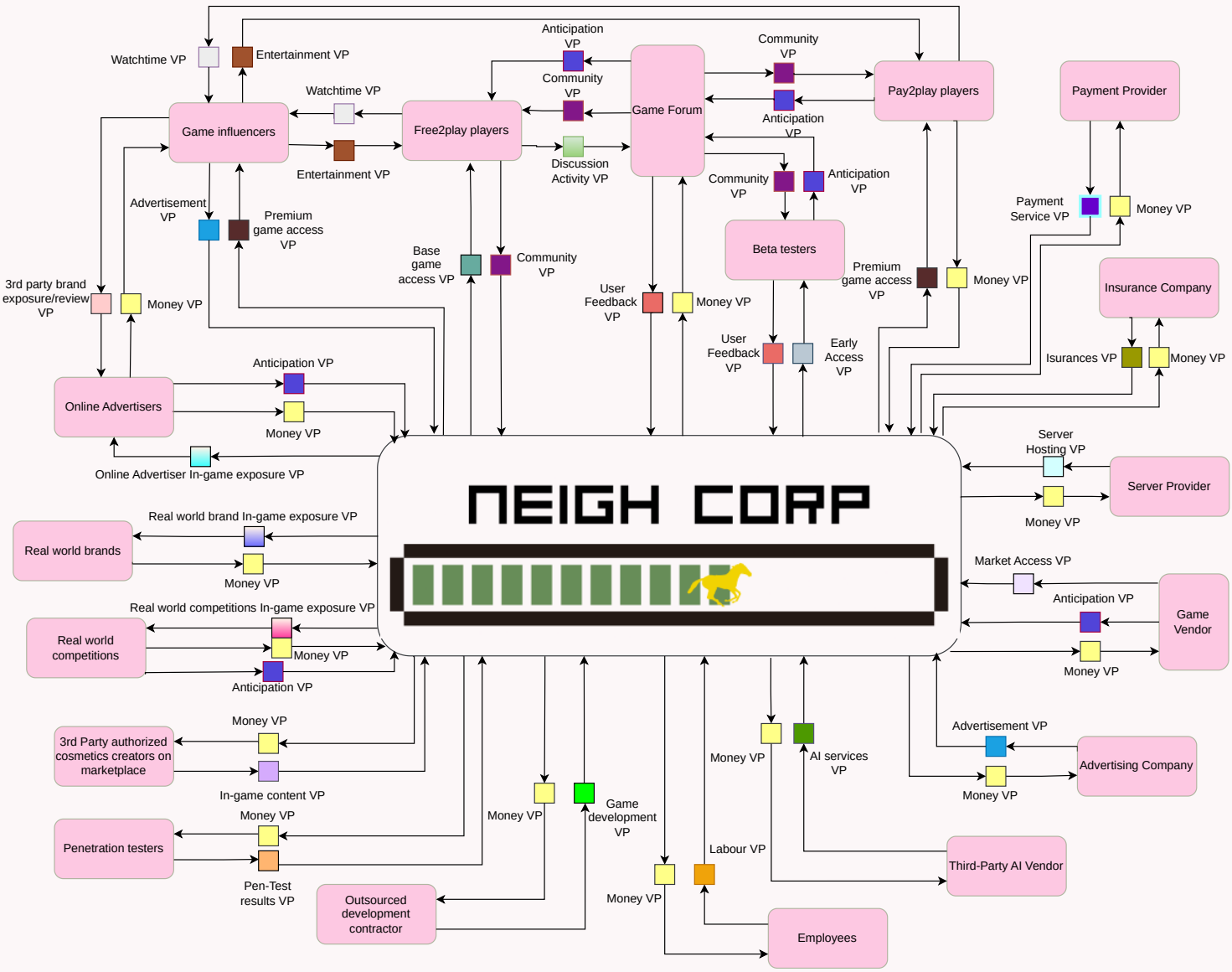


Figure 1: VDML Diagram of *Neigh Corp*

2.2 VDML Descriptions

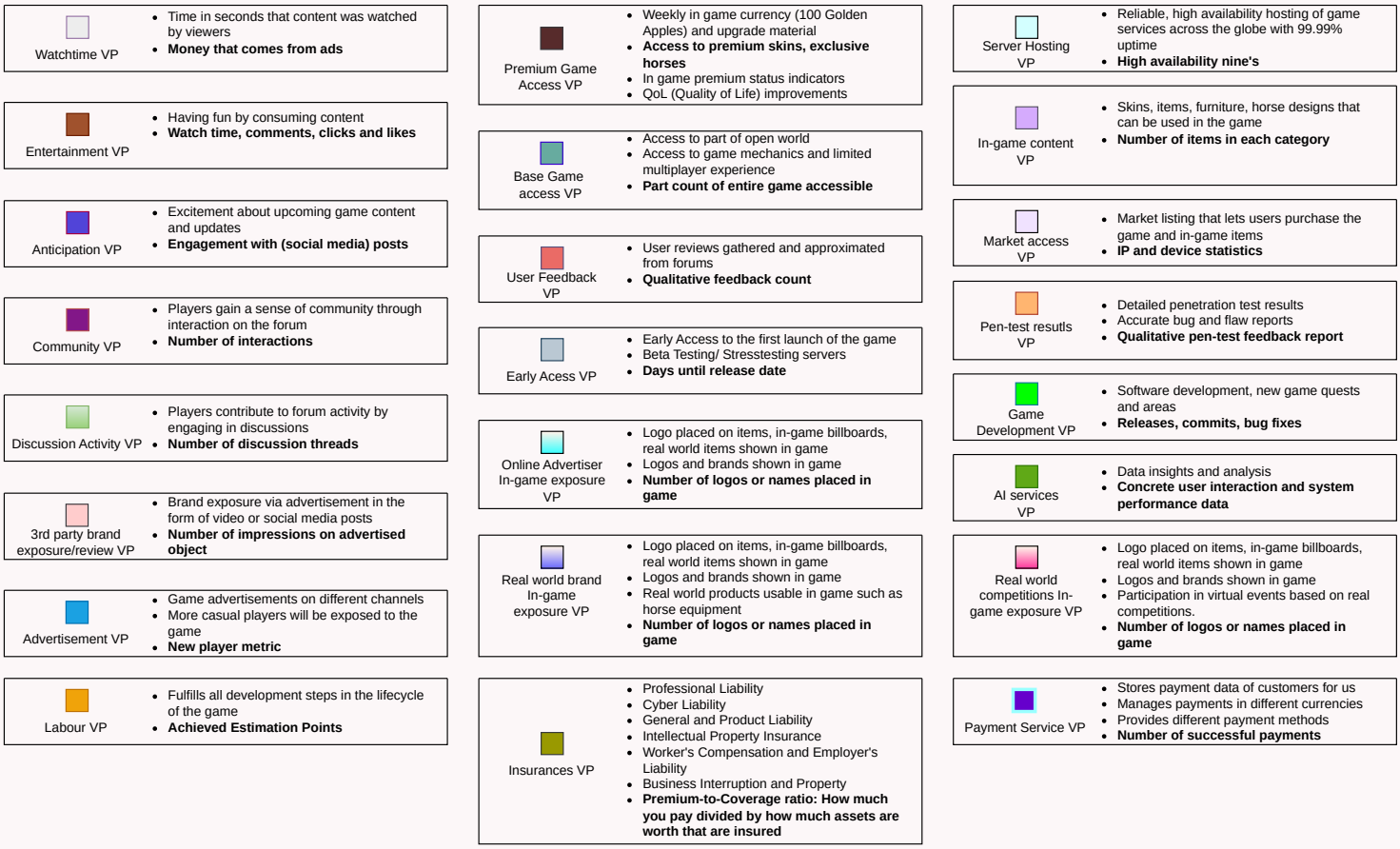


Figure 2: VDML Descriptions of *Neigh Corp*

2.3 VDML Diagram Explained

Seen above is the VDML diagram designed for Golden Meadows, featuring Neigh Corp as the central agent involved in the majority of value exchange processes. The diagram provides a comprehensive overview of the majority of actors involved in value propositions directly affecting the game.

The inclusion of individual agents was decided based on the previously-designed goal model as well as the actors typically engaged in the game development as well as MMO game maintenance processes. Each of the identified agents' involvements in a value proposition is briefly described below, as well as the agent itself where its name is not self-explanatory.

2.3.1 Value Propositions between External Agents

Several actors included on the VDML diagram are engaged in value exchanges without Neigh Corp's direct involvement, namely:

1. **Free-to-play players as well as pay-to-play players and game forum**

   The *Free-to-Play Players* actor, representing users of *Golden Meadows* who do not conduct any in-game purchases, is engaged in a value exchange with *Game Forum*, representing the online community where players can participate in conversations via forum posts. *Pay-to-Play Players*, who are denoted as users actively purchasing in-game items and premium benefits, are, too, engaged in an identical value exchange with the exact same VPs. The following value propositions are involved in this relationship:

   (a) Community VP and Anticipation VP - given to the *Players*;

   (b) Discussion Activity VP - given to the *Game Forum*.

2. **Game influencers and Free-to-Play as well as Pay-to-Play Players**

   The *Free-to-Play* and *Pay-to-Play Player* actors are also engaged in another value exchange with the *Game Influencers* - an actor representing agents such as popular players, You-Tubers, as well as streamers contributing to the game's popularity. The value propositions are:

   (a) *Watch Time VP* - given to the *Influencers*;

   (b) *Entertainment VP* - given to the *Players*.

3. **Game Influencers and Online Advertisers**

   The value exchange between **Game Influencers** and **Online Advertisers** involves the following value propositions:

   (a) Money VP - given to *Game Influencers*;

   (b) Third-party Brand Exposure/Review VP - given to the *Online Advertisers*.

2.3.2 Value Propositions involving Neigh Corp

The majority of value exchange relationships depicted on the VDML diagram include Neigh Corp, described below:

1. **Pay-to-play players**

   The exchange between pay-to-play players and Neigh Corp involves *Money VP*, given to Neigh Corp, as well as *Premium Game Access VP*, given to the pay-to-play players.

2. **Game Forum**

The *Game Forum* agent receives the *Money VP* from Neigh Corp in exchange for *User Feedback VP*.

3. **Free-to-play players**

   Neigh Corp receives *Feedback VP* from the Players agent, defined on the diagram seen above, in exchange for *Base Game Access VP*.

4. **Beta testers**

   Beta Testers receive *Early Access VP*, and in turn, Neigh Corp receives *User Feedback VP*. Both value propositions are given concrete definitions on the image seen above, below the VDML diagram.

5. **Game Influencers**

   Game Influencers receive *Premium Game Access VP* in exchange for *Advertisement VP*.

6. **Online Advertisers**

   Online Advertisers are engaged in a value exchange with Neigh Corp, receiving the *Online Advertiser In-game exposure VP*. The actor, in turn, gives *Money VP* as well as *Anticipation VP*.

7. **Real-world brands**

   Real-world brands provide Neigh Corp with the *Money VP*. The actor receives *Real world competitions In-game exposure VP* (see definition).

   The *Real-world brands* actor represents real-life companies wishing to engage in a partnership with Neigh Corp, provided with ad placement within the game in exchange for a monthly or one-time fee, decided on a case-by-case basis and the duration of ad placement.

8. **Third-party authorized cosmetics creators on marketplace**

   The actor represents individual users or groups of people engaged in the development of in-game cosmetics. In exchange for the provided *In-Game Content VP*, the actor receives *Money VP* from Neigh Corp.

9. **Penetration testers**

   Penetration testers receive *Money VP* from Neigh Corp in exchange for conducting penetration tests in order to identify potential security vulnerabilities within the company's infrastructure, represented by the *Pen-Test Results VP*.

10. **Outsourced Development Contractor**

    Third-party developers hired by Neigh Corp for various development tasks are represented by this agent, receiving the *Money VP* in exchange for *Game Development VP*.

11. **Third-Party AI Vendor**

    Third-Party AI Vendors is an agent representing companies providing Neigh Corp with a miscellaneous set of AI tools, for instance, copilots, agents, or data gathering software. The provided *AI Services VP* is exchanged for *Money VP*.

12. **Advertising Company**

    The advertising company actor receives *Money VP* from Neigh Corp in exchange for the advertising services the agent provides, represented by the *Advertisement VP*.

13. **Game Vendor**

    The *Game Vendor* agent, denoting a game marketplace providing companies with a platform to sell their product on, exchanges *Anticipation VP* for *Money VP*, anticipation in this context meaning teasers being released ahead of the game's or its update's release.

14. **Server Provider**

    The *Server Provider* agent gives Neigh Corp the *Server Hosting VP*, in turn receiving *Money VP*.

15. **Insurance Company**

    Insurance Company provides Neigh Corp with the *Insurances VP*, which represents one or more active insurance agreements. Neigh Corp pays for said insurance with the *Money VP*.

16. **Employees**

    The agent representing *Employees* receives *Money VP* from the company in exchange for *Labour VP*.

17. **Payment Provider**

    The agent representing *Payment Providers* receives *Money VP* from the company in exchange for *Payment Service VP*.

3 Goal Design

This section is to show the goals of the company. The goals shall be clearly related to the value network analysis in Section 1. In order to identify relevant goals, the project group shall make use of the five forces model and generic strategies of Michael Porter (see links on the page "Reading Resources" at the course web site). It is recommended that Strategy maps be used to structure the goal model (see link on the same page).

3.1 Goal Model

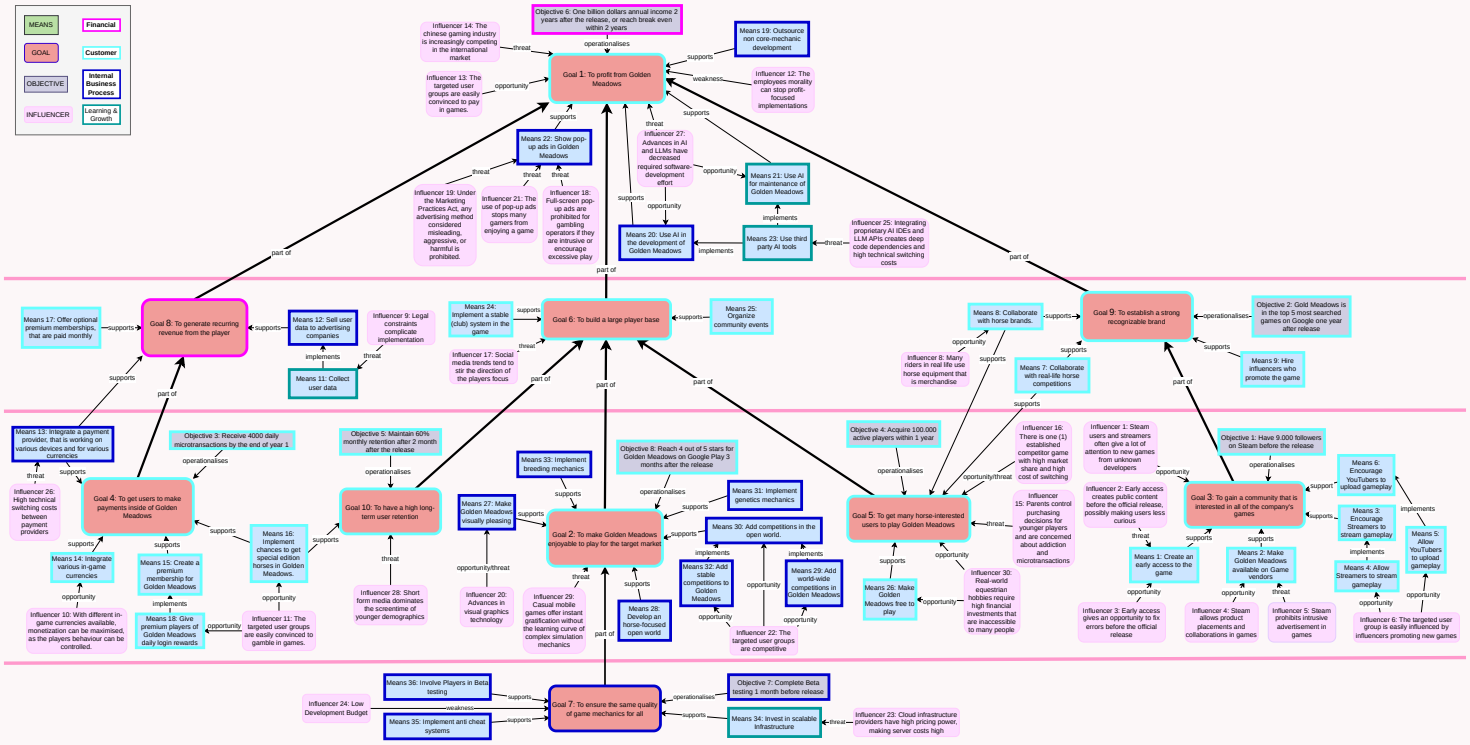


Figure 3: Goal model for *Neigh Corp*

3.2 Goal Model Explained

Figure 3 shows the goal model for Neigh Corp. It describes the overarching goals, subgoals, influencers that affect these goals, and the means that can help achieve them. The coloured borders around the boxes in the diagram, as well as the horizontal pink lines dividing the model into four tiers, indicate which dimension of value creation in the company these belong to, in accordance with Kaplan and Norton's Strategy Maps [1]. Which colour corresponds to which level is shown in the diagram next to the legend. The levels shown in the diagram are ordered in accordance with the perspectives in the Strategy Maps, ranging from financial goals at the top to concrete technologies and infrastructure at the bottom. Influencers are modelled outside any perspective, as these can impact across several categories and are not controlled by the company.

The first level at the bottom, named "Learning & Growth", contains the foundational technologies, tools, and infrastructure used in the company. This level supports the one above it, "Internal Business Processes", which lists the internal development methods and operational processes the company uses to create

value. The "Customer" perspective contains the most elements and describes how these internal processes create differentiated value and loyalty for the customer, and the final "Financial" level at the top shows how all these elements combine to fulfil the most important monetary goals in the company, such as profit and recurring revenue.

The goal model was created with the previously shown Figure 1 diagram in mind and in accordance with Porter's five forces [2] and Porter's generic strategies. Each of Porter's five forces is listed below with an analysis of which parts of the diagram belong to it, which generic strategy is used to combat its force, and the reasoning behind their Strategy Maps classification. The overarching approach for Golden Meadows is differentiation focus, as the target customer group for this game is rather niche.

Competitive rivalry: This force measures how many competing products there are on the market. A large number of competitors results in a greater force. Influencers 14 and 16 measure this force. They indicate that there is increasing competition from Chinese gaming companies and that there is one established competitor with high market share and a high cost of switching. This makes the competitive rivalry force comparatively strong, since the potential for new competitors emerging has increased, and potential customers experience a high cost of switching to Golden Meadows.

A number of goals and means are in place to combat this force. Means 26 is about making Golden Meadows free to play, as shown in the VDML Figure 1 with the Free-to-Play Players actor. This makes the game accessible to new users at no cost and therefore lowers the upfront switching cost to zero. Means 26 belongs to the "Customer" perspective, as it directly shapes the value proposition offered to players and supports Goal 5, getting many horse-interested users to play Golden Meadows (operationalised by Objective 4).

Goal 9, establishing a strongly recognizable brand (operationalised by Objective 2), also combats this force by establishing Golden Meadows' individual identity. While competitors can copy graphics, the means that go into this goal, such as collaborating with real-life horse competitions and horse brands (Means 7 and 8, capitalising on Influencer 8) and hiring influencers (Means 9), are harder to replicate. Both Goal 9 and Means 7, 8, and 9 are placed in the "Customer" perspective because they focus on external brand recognition and customer perception. These means and goals relate directly to the VDML Game Influencers and Real-world brands actors that provide Neigh Corp with advertisement and in-game brand exposure. In the Strategy Map, Goal 5 and Goal 9 support Goal 6 (building a large player base in the "Customer" perspective), which in turn drives the top-level "Financial" goal, Goal 1 (to profit from Golden Meadows).

Bargaining Power of Buyers: This force measures how much power the buyers have. The influencers in the goal model that show weakness in this force are Influencers 11 and 13, which state that the target group is easily convinced to gamble and pay in games. The factors that strengthen this force are Influencers

15 and 21, which state that parents control the purchases of younger players and that pop-up ads can decrease their game enjoyment.

To combat this force, among others, Means 17 is implemented, which presents the customers with an optional, regular subscription for continued premium game access. This can appease the minds of concerned parents with a predictable monthly expense. In the Strategy Map, Means 17, alongside Means 14, 15, 16, and 18 (integrating various in-game currencies, creating premium memberships, offering gacha chances for special edition horses, and giving daily login rewards) belongs to the "Customer" perspective, as these elements define the purchasing options and reward structures experienced by the player. They support Goal 4 (getting users to make in-game payments, operationalised by Objective 3) and Goal 10 (high long-term user retention, operationalised by Objective 5) in the "Customer" perspective, which then flow upward into the "Financial" perspective to fulfil Goal 8 (generating recurring revenue) and Goal 1 (profiting from Golden Meadows). Conversely, Means 22 (showing pop-up ads) belongs to the "Internal Business Process" perspective and is threatened by Influencer 21, as intrusive internal ad implementations reduce player enjoyment. This subscription and payment structure is directly related to the VDML value transfer between Neigh Corp and Pay-to-Play Players. Buyer power is further weakened by the niche approach that differentiates Golden Meadows from competitors. Therefore, customers have few similar alternatives to switch to and their bargaining power is reduced.

Bargaining Power of Suppliers: This force measures how much power the suppliers have over the company in terms of pricing, platform rules, and technical dependencies. In the goal model, this force is strengthened by Influencer 5, which states that Steam prohibits intrusive advertisements in games, and Influencer 23, which states that cloud infrastructure providers have high pricing power, making server costs high. It is further strengthened by Influencers 25 and 26, which highlight that integrating proprietary AI tools and payment providers creates deep code dependencies and high technical switching costs. On the other hand, the force is slightly weakened by Influencer 4, which notes that Steam allows product placements and collaborations within games.

To address supplier power across the Strategy Map, technological dependencies are handled in the foundational "Learning & Growth" and "Internal Business Process" perspectives. Means 34 (investing in scalable infrastructure, threatened by Influencer 23) and Means 23 (using third-party AI tools, threatened by Influencer 25) belong to the "Learning & Growth" perspective because they represent the core IT infrastructure and external software capabilities of the company. Means 34 supports Goal 7 (ensuring the same quality of game mechanics for all) in the "Internal Business Process" perspective, while Means 23 implements Means 21 (using AI for maintenance in "Learning & Growth") and Means 20 (using AI in development in "Internal Business Process"). Similarly, Means 13 (integrating a multi-device, multi-currency payment provider, threatened by Influencer 26) is classified as an "Internal Business Process" that

enables Goal 4 in the "Customer" perspective. Because the Game Vendor's restrictions on intrusive ads (Influencer 5) limit the use of pop-up advertisements (Means 22) on Steam, Neigh Corp complements its ad model with "Customer" perspective initiatives that take advantage of Influencer 4, such as making the game available on game vendors (Means 2) and collaborating with real-life horse competitions and brands (Means 7 and 8). In the VDML diagram, these relationships correspond to the Server Provider, Third-Party AI Vendor, Payment Provider, Game Vendor, and Real-world brands actors. By using AI to lower development costs and establishing steady recurring revenue (Goal 8) in the "Financial" perspective, Neigh Corp is able to absorb the high costs and switching barriers imposed by its technical suppliers.

Threat of New Entrants: This force measures how easily new competitors can enter the market and compete with Golden Meadows. Influencer 27 strengthens this force by stating that advances in AI and LLMs have decreased the required software-development effort, which lowers the barrier to entry for new studios. Influencer 20 (advances in visual graphics technology) also strengthens this threat against Goal 2, as it raises players' expectations for graphical fidelity and allows new entrants with modern tools to attract attention. Furthermore, Influencer 1 indicates that Steam users and streamers often give a lot of attention to new games from unknown developers; while this is an opportunity for Neigh Corp at launch, it also makes market entry easier for future newcomers.

Neigh Corp combats the threat of new entrants by turning AI into an internal advantage and building complex, niche game mechanics that cannot be quickly replicated. In the Strategy Map, Neigh Corp uses Means 20, 21, and 23 (spanning the "Learning & Growth" and "Internal Business Process" perspectives) alongside Means 19 (outsourcing non-core mechanic development in the "Internal Business Process" perspective) to reduce its own development costs and directly support Goal 1 in the "Financial" perspective. To defend against rising graphical expectations (Influencer 20) and create a deep product barrier, the company implements several "Internal Business Process" means: making the game visually pleasing (Means 27), developing a horse-focused open world (Means 28), adding world-wide, open-world, and stable competitions (Means 29, 30, and 32), and implementing genetics and breeding mechanics (Means 31 and 33). These are placed in the "Internal Business Process" perspective because they represent core development tasks carried out by Neigh Corp's employees and Outsourced Development Contractors in the VDML. Moving up the Strategy Map, these internal processes support Goal 2 (making Golden Meadows enjoyable to play for the target market, operationalised by Objective 8) in the "Customer" perspective. Simultaneously, Neigh Corp leverages Influencer 1 through "Customer" perspective means: creating early access (Means 1), publishing on game vendors (Means 2), and allowing and encouraging streamers and YouTubers to share gameplay (Means 3, 4, 5, and 6) to achieve Goal 3 (gaining a loyal community, operationalised by Objective 1). This connects to the Beta Testers, Game Vendor, and Game Influencers in the VDML and establishes an

early player base before new entrants can establish themselves.

Threat of Substitutes: This force measures how easily customers can satisfy their needs with alternative products or activities outside of the direct horse-game niche. In the goal model, Influencers 28 and 29 strengthen this force: Influencer 28 states that short-form media dominates the screentime of younger demographics (threatening Goal 10), and Influencer 29 states that casual mobile games offer instant gratification without the learning curve of complex simulation mechanics (threatening Goal 2). However, Influencer 30 states that real-world equestrian hobbies require high financial investments that are inaccessible to many people. This is beneficial for Neigh Corp, as it weakens the threat of the primary real-life substitute and acts as an opportunity for Goal 5.

To capitalise on Influencer 30 and counter Influencer 29, Neigh Corp combines its "Internal Business Process" simulation mechanics (Means 28–33) with the "Customer" perspective Means 26 (making Golden Meadows free to play). Because real-life horse riding is prohibitively expensive, offering an immersive, free-to-play virtual alternative in the "Customer" perspective makes Golden Meadows highly attractive to horse-interested users (Goal 5), serving both Free-to-Play and Pay-to-Play Players in the VDML. To combat the distraction of short-form media (Influencer 28) and secure high long-term user retention (Goal 10 in the "Customer" perspective), Neigh Corp implements social and reward-based means in the "Customer" perspective: implementing a stable/club system (Means 24), organizing community events (Means 25), offering special edition horses (Means 16), and providing daily login rewards (Means 18). These social features relate to the Game Forum actor in the VDML and support Goal 6 (building a large player base) and Goal 10 in the "Customer" perspective, which subsequently secure steady in-game payments (Goal 4) and recurring revenue (Goal 8) in the "Financial" perspective.

Generic strategies: To combat all five forces in a single, consistent way, Neigh Corp applies a differentiation focus (niche) strategy paired with a grassroots, influencer- and community-driven market approach. Rather than competing with massive studios for a general gaming audience, every level of the Strategy Map is aligned toward serving the specific niche of horse enthusiasts.

At the bottom of the Strategy Map, the "Learning & Growth" and "Internal Business Process" perspectives focus on cost-effective scalability and niche quality. By using AI tools (Means 20, 21, 23), outsourcing non-core tasks (Means 19), and investing in scalable infrastructure, anti-cheat systems, and beta testing (Means 34, 35, 36 supporting Goal 7), Neigh Corp keeps its operational costs under control despite a low development budget (Influencer 24) and strong supplier power (Influencers 23, 25, 26). This allows the internal development team to focus on specialised equestrian simulation mechanics (Means 27–33).

In the "Customer" perspective, these specialised internal processes translate into a differentiated product (Goal 2) that protects Neigh Corp against competitive rivalry, new entrants, and casual substitutes. Although showing pop-up

ads (Means 22) is included as a supplementary revenue stream, Neigh Corp cannot rely heavily on them because intrusive advertisements alienate players and face strict legal and platform constraints (Influencers 5, 18, 19, 21). Instead, Neigh Corp primarily drives growth through a grassroots marketing strategy. By encouraging streamers and YouTubers (Means 3–6, 9), offering early access (Means 1), fostering clubs and community events (Means 24, 25), and partnering with real-life equestrian brands and competitions (Means 7, 8), the company builds an authentic brand (Goal 9) and an engaged community (Goals 3, 5, 6, 10) across the VDML network. Finally, at the top of the Strategy Map, this customer loyalty and niche differentiation weaken buyer power and convert directly into sustainable microtransactions (Goal 4) and recurring monthly subscriptions (Means 17, Goal 8), fulfilling the overarching "Financial" goal of profiting from Golden Meadows (Goal 1, Objective 6).

4 Process Design

This section is to show the processes of the company. The processes shall be clearly related to the value network analysis in Section 2 and the goal design in Section 3.

4.1 Exchange and Conversion Processes in REA

4.1.1 Conversion Processes

Seen above is the diagram depicting every conversion process Neigh Corp is involved in. The conversion processes included are as follows:

1. **Merch Design Conversion Process**

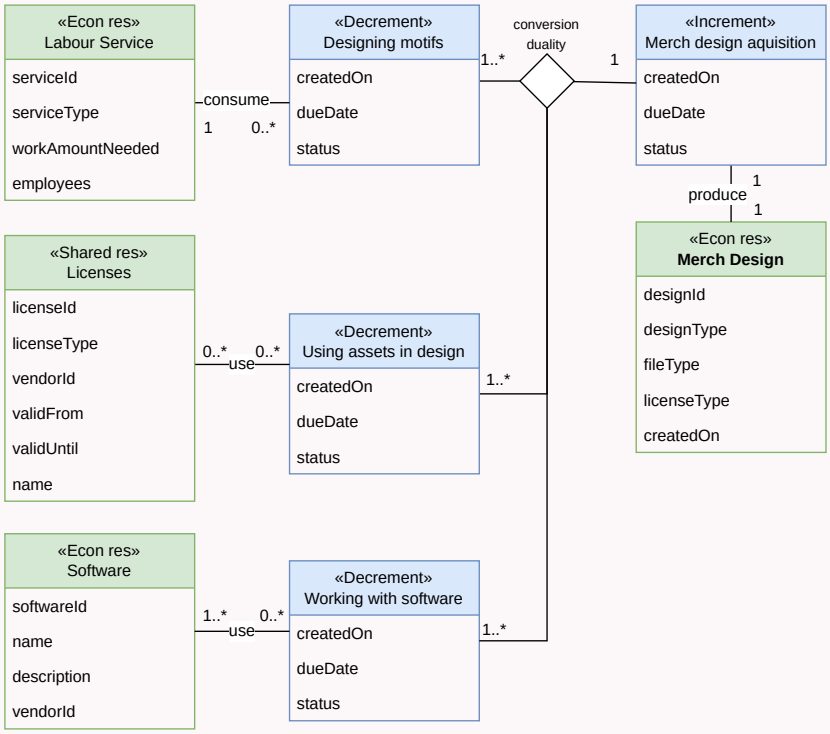


Figure 4: Merch Design Conversion Process

The economic event *Designing motifs* decrements the *Labour Service* resource with a 1 to 0..* multiplicity, meaning that one and only one labour service, representing the act of employees working, can simultaneously be engaged in many motif designs, or as few as zero. The resource is **consumed**.

Working with Software decrements software in a **use** relationship, i.e., the software is not permanently destroyed in the process and can be used again in future processes. The software may be used by zero or many of these events, and the event itself may be using one or many software resources, as the event name itself implies at least one software resource has to be used.

Lastly, the event *Using assets in design* decrements *Licenses* - the resource representing licenses that may be necessary to acquire in order to use certain design elements (assets, as the event name implies). Zero or many licenses may be necessary as there is no guarantee copyrighted material will be used. For that reason, licenses are also engaged in zero-to-many of

these events, as they may or may not be used by many asset design use events.

Finally, each event is engaged in a one-to-many relationship with the conversion process, as each of the events may need to occur multiple times to finalise a merchandise design. The events are converted into the *Merch Design Acquisition* increment event, which produces the final **Merch Design** resource.

2. Character Design Conversion Process

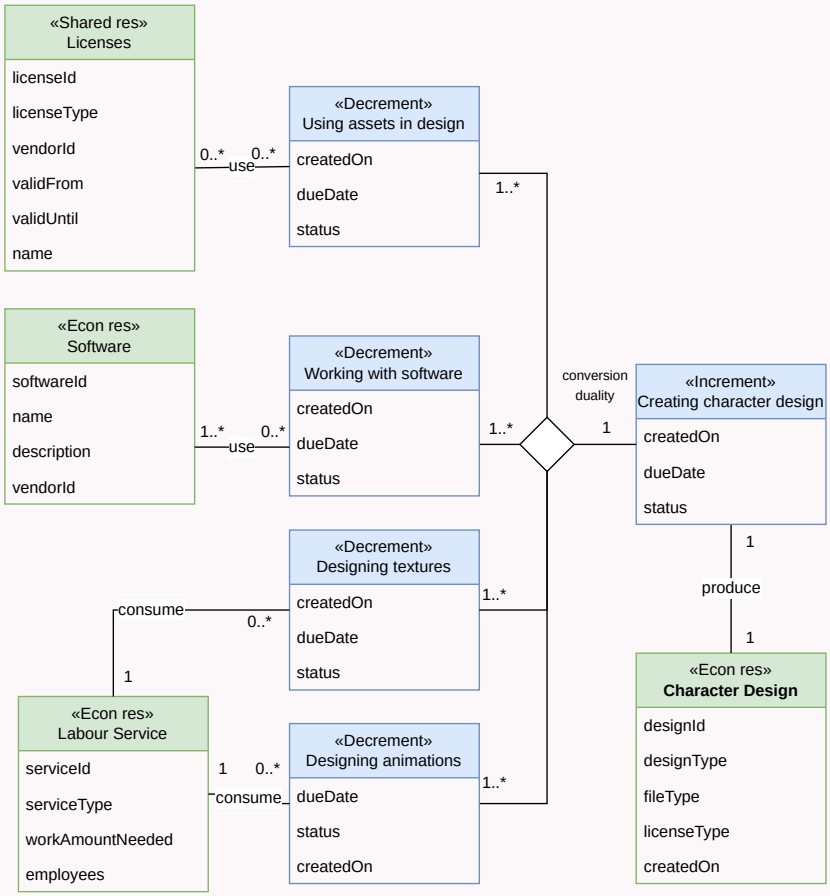


Figure 5: Character Design Conversion Process

The same two resources described in the previous process are engaged in this conversion process as well, using identical multiplicities, with the exception of *software*.

Labour Service is consumed in a 1 to 0..* relationship with both the *Designing Textures* and *Designing Animations* decrement events. The labour service, i.e. the employees, may not necessarily be engaged in either of these events, hence the multiplicity of 0..*, but if these events are taking place, labour service must always be consumed by both of them. The *Working with software* decrement event takes place once again, engaged in the same **use** relationship with *software* as before. The three decrement events are converted into the *Creating Character Design* increment event, which produces **Character Design**.

3. Merch Conversion Process

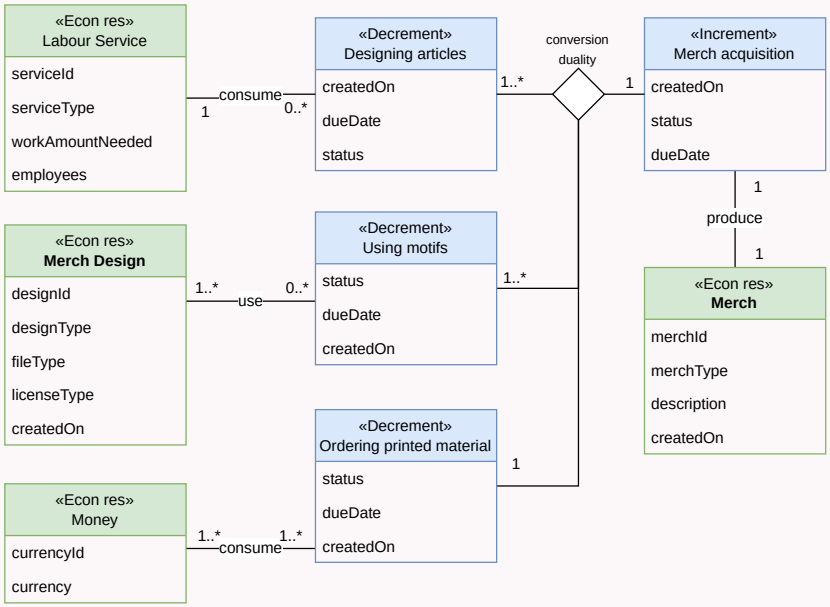


Figure 6: Merch Conversion Process

The corresponding decrement events of the process are as follows:

Designing Articles - the event consumes *Labour Service* in a 1 to 0..* relationship, the multiplicity reasoning being identical to what was discussed in the previous conversion processes regarding said resource.

The *Merch Design* resource produced by the first conversion process is used by the *Using motifs* economic event in a 1..* to 0..* relationship, meaning that one or many merchandise designs may be used by many of these events at once, but they don't have to be, either because there is no event taking place or because the merchandise design simply remains unused. If, however, the event is happening, at least one motif has to be used.

Lastly, *Money* economic resource is consumed by the *Ordering printed material* event in a 1..* to 1..* relationship.

Finally, the three economic events are converted into the *Merch Acquisition* event which produces the **Merch** resource.

4. **High-quality Game Content Conversion Process**

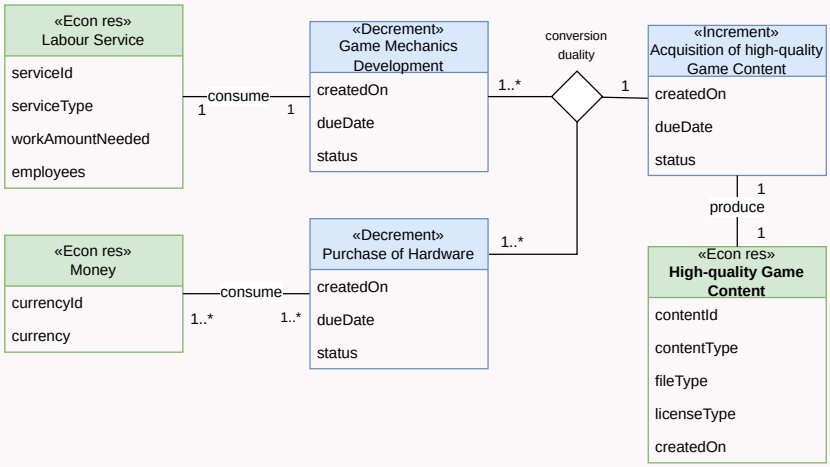


Figure 7: High-quality Game Content Conversion Process

Two economic decrement events are engaged in this conversion process, the first one being *Game Mechanics Development*, which consumes the *Labour Service* resources in a 1 to 1 relationship, meaning that the employees represented by the resource are always working on a new game mechanic, and if they are, they are always working on only one.

The second event is *Purchase of Hardware*, which consumes *Money* in a 1..* to 1..* relationship (i.e., one purchase requires an abstract unit of money, and the same unit of money may be used on many purchases).

The two events are converted into the *Acquisition of High-Quality Game Content* event, which produces the **High-Quality Game Content** resource.

5. **Brand Awareness Conversion Process**

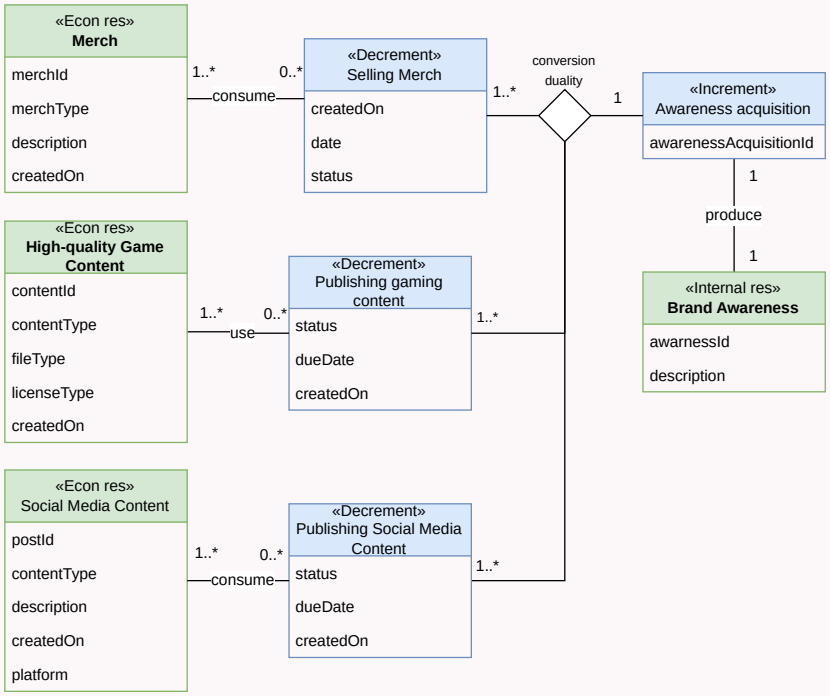


Figure 8: Brand Awareness Conversion Process

Three economic events participate in this conversion process. The first decrement event, *Selling Merch*, consumes the *Merch* resource produced in the earlier conversion process in a 1..* to 0..* relationship, meaning that if merchandise is being sold, at least one unit of merchandise must actually be consumed in the process, though not limited to one. Merchandise itself does not necessarily have to be in the process of being sold, and it can be sold by multiple *Selling Merch* events given that more copies can be theoretically made.

Moving on, the *Publishing Game content* decrement **uses** the *High-Quality Game Content* produced in the previous process, the multiplicities being 1..* to 0..*. This implies that at least one unit of high-quality game content must be used by the event, as its very name suggests, though there is no set limit. Meanwhile, *High-Quality Game Content* can exist as a unit without having to be actively used, hence the 0..* on the right side.

The last economic event is *Publishing Social Media Content*, which consumes the *Social Media Content*, the reasoning being that once a social media post is made public, publishing it again will be redundant, as it will

have already used up its value as a form of advertisement. The multiplicities between the resource and the event are 1..* to 0..*, using roughly the same logic as in the previous resource-event relationship (the event name implies at least one resource has to be used for the event to occur, and the resource can exist without being engaged in the event).

The three events are converted into the *Awareness Acquisition* increment event, which, finally, produces **Brand Awareness**, representing how well-known Neigh Corp's company name is, as well as the game itself.

4.1.2 Exchange Processes

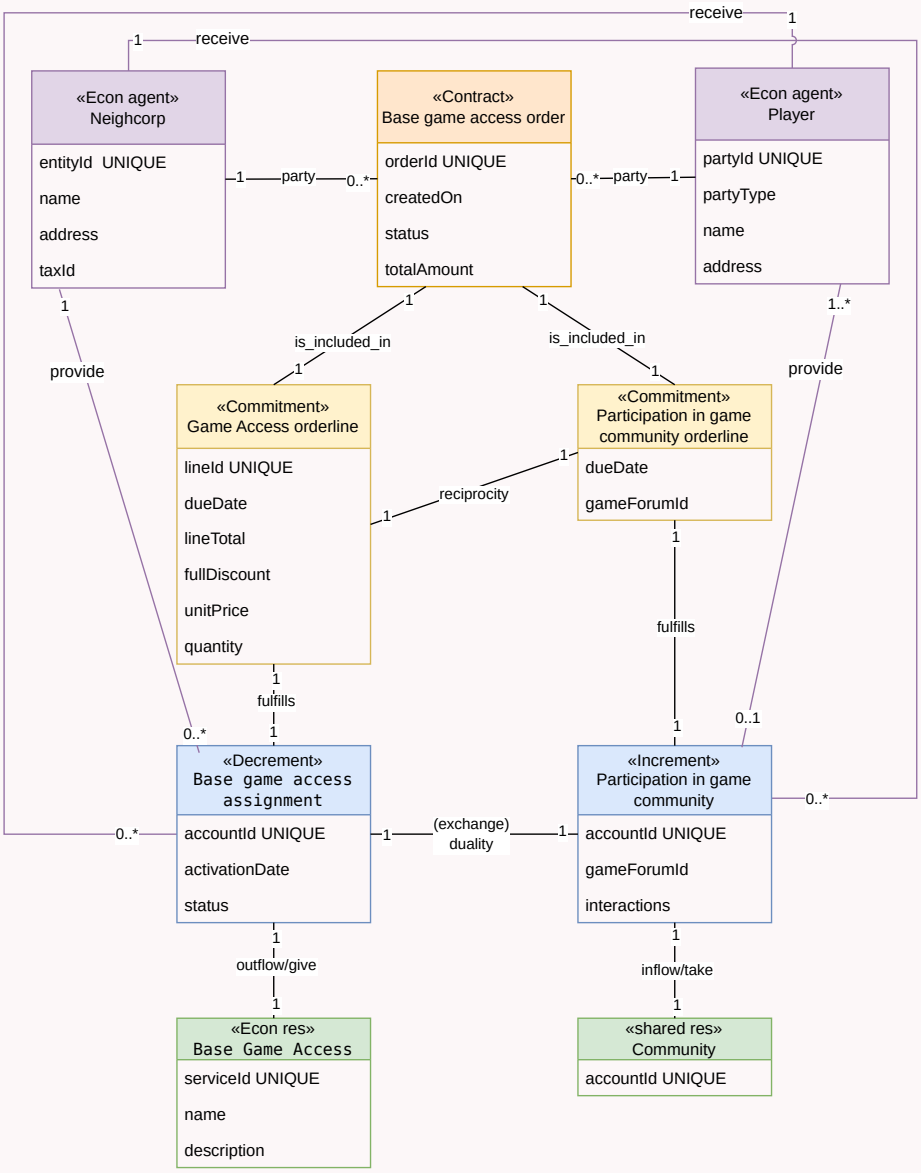


Figure 9: REA Base Game Access Exchange

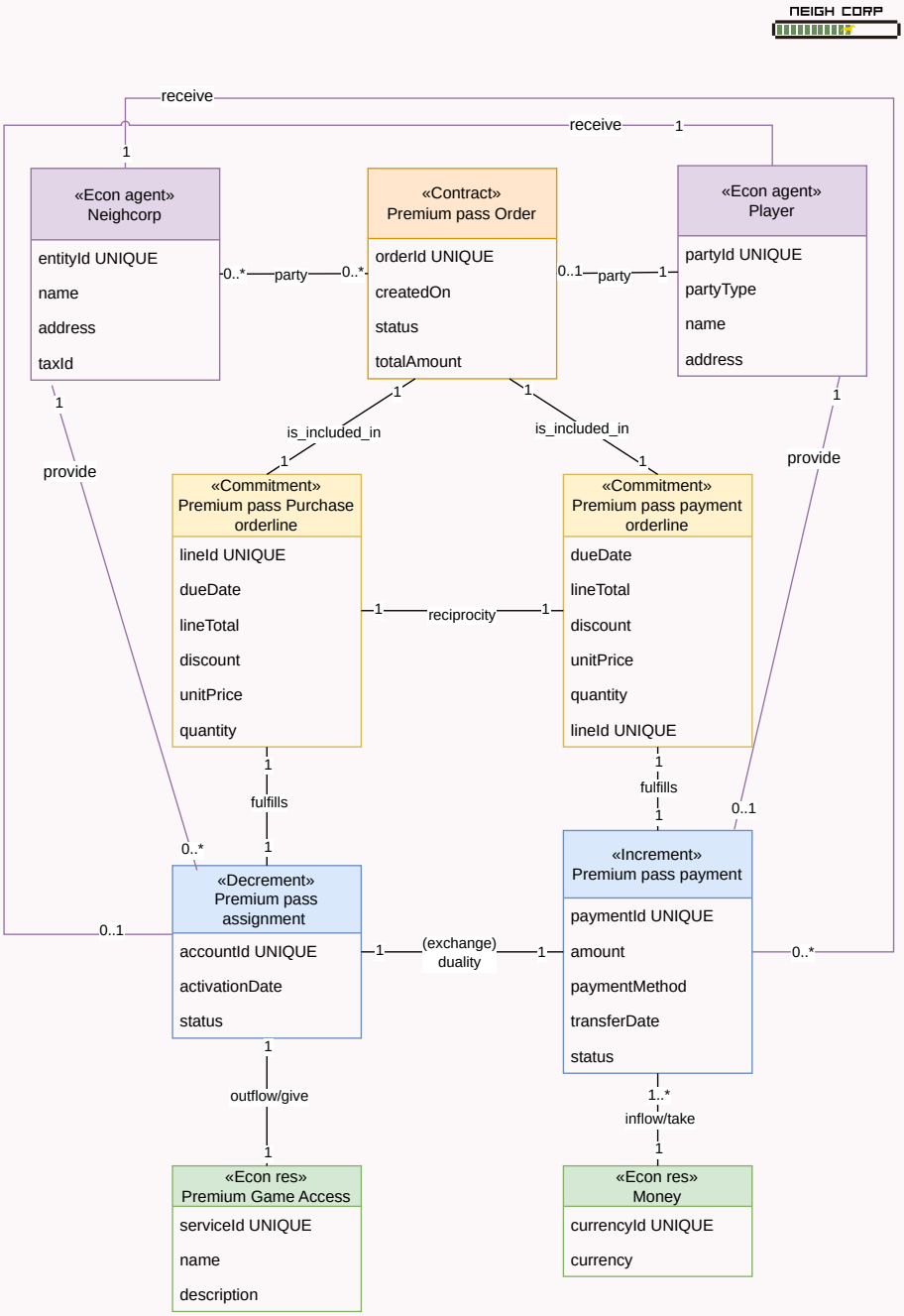


Figure 10: REA Premium Game Acess

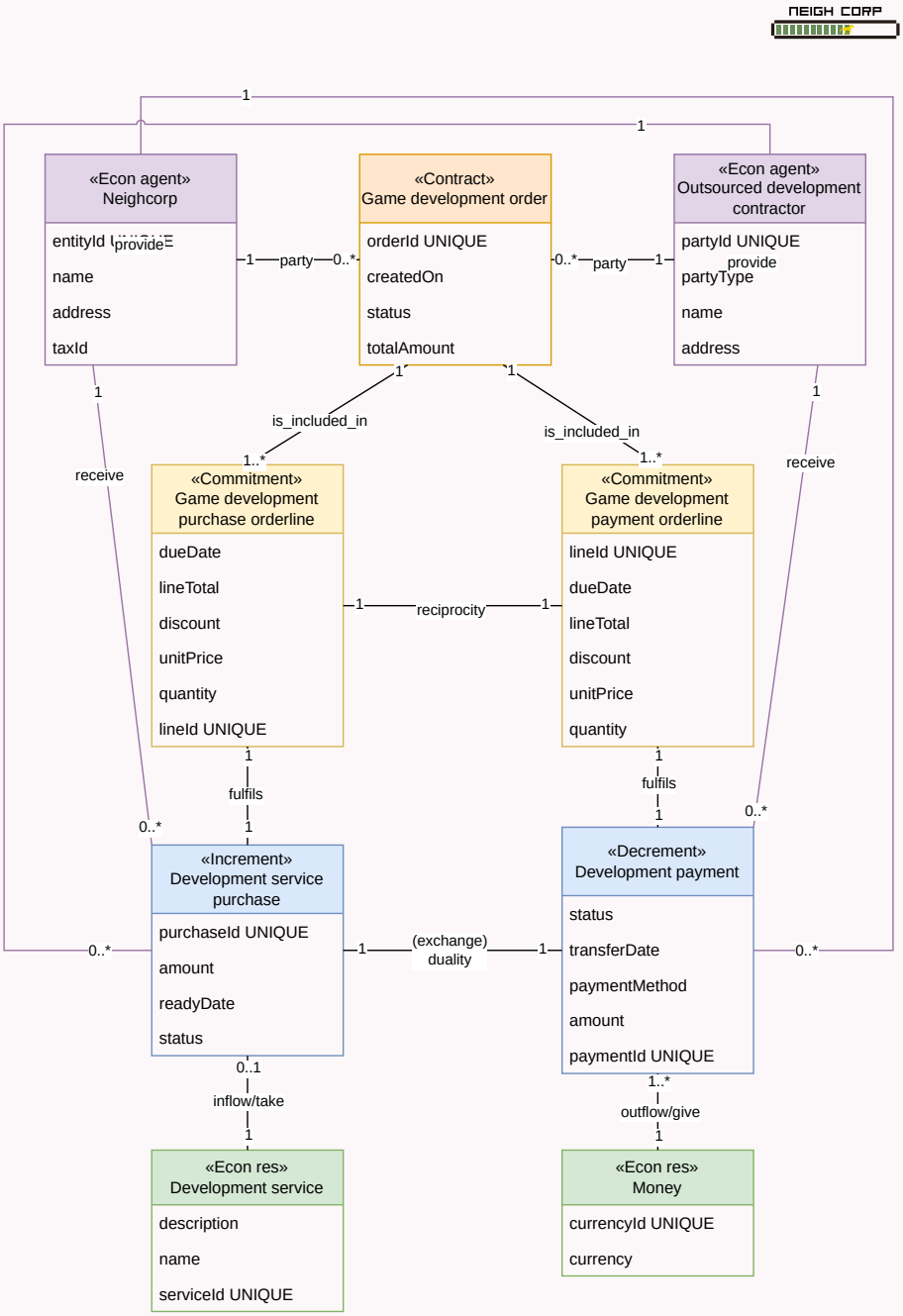


Figure 11: REA Game Development Exchange

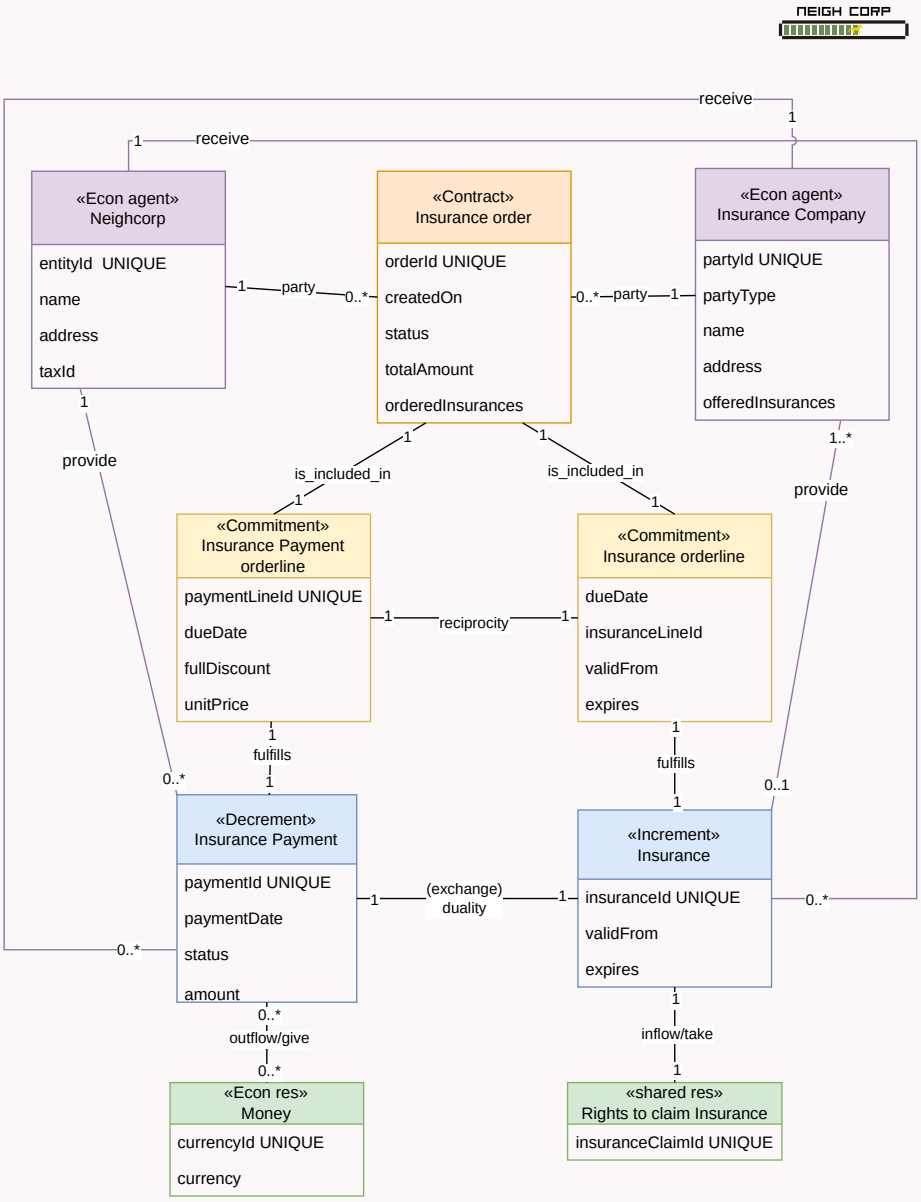


Figure 12: REA Insurance Exchange

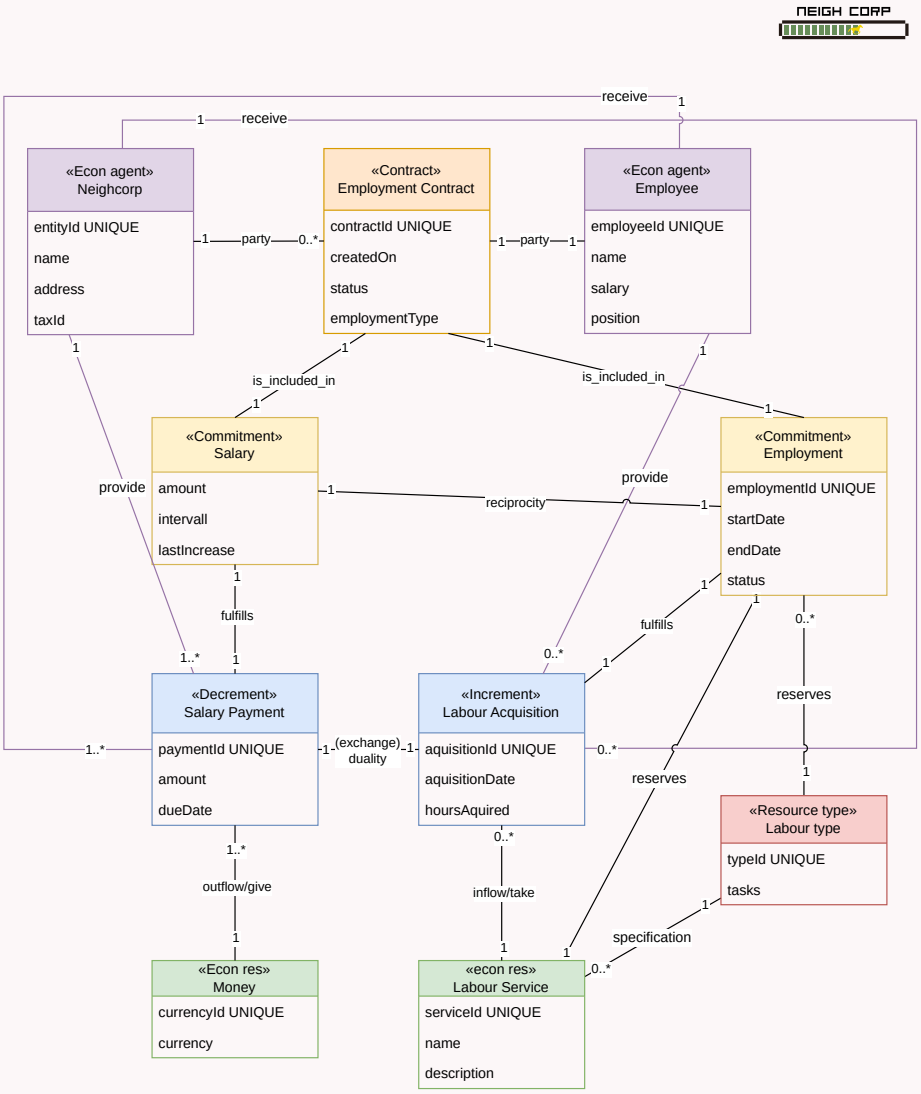


Figure 13: REA Labour Service Exchange

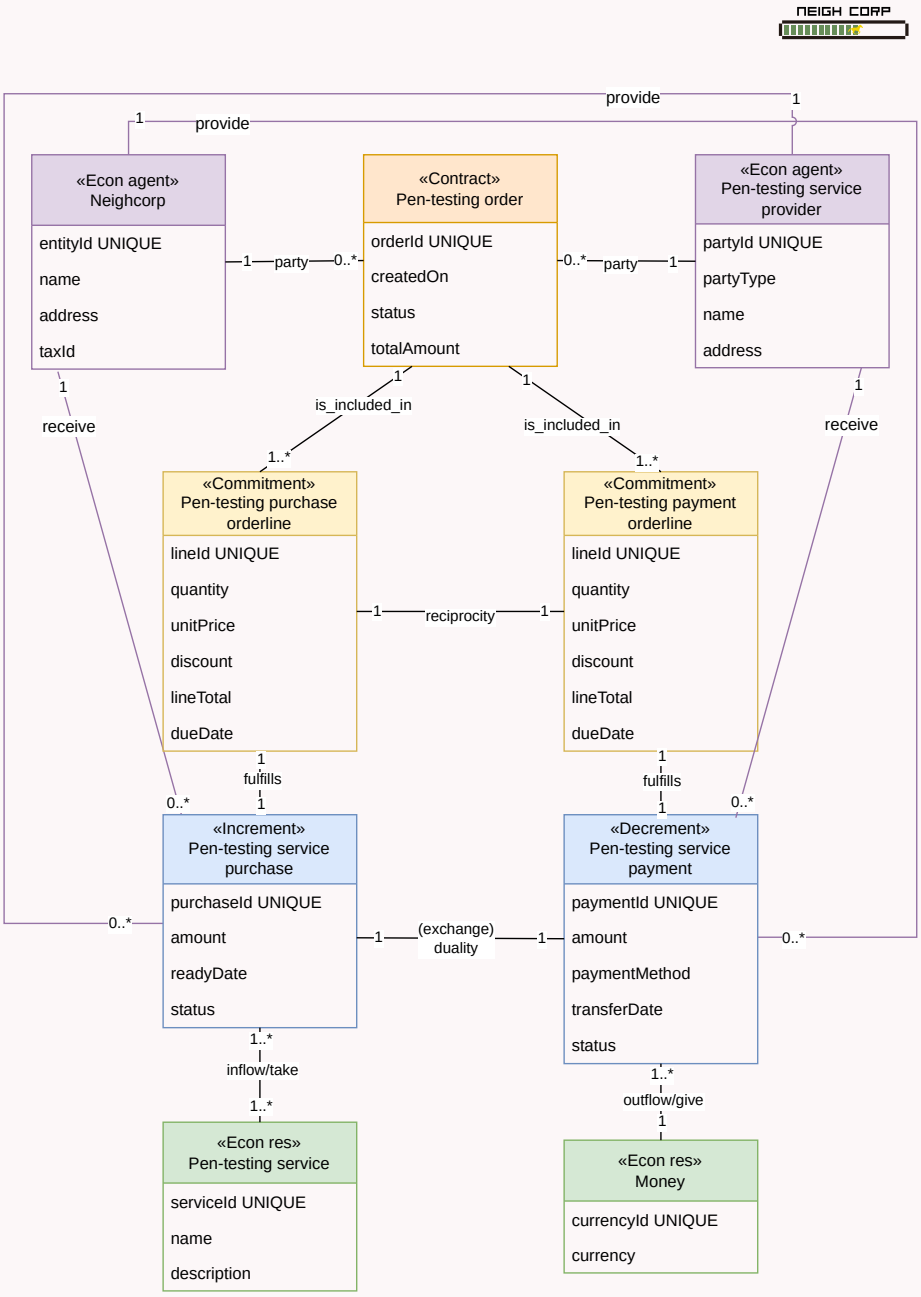


Figure 14: REA Pentest Service Exchange

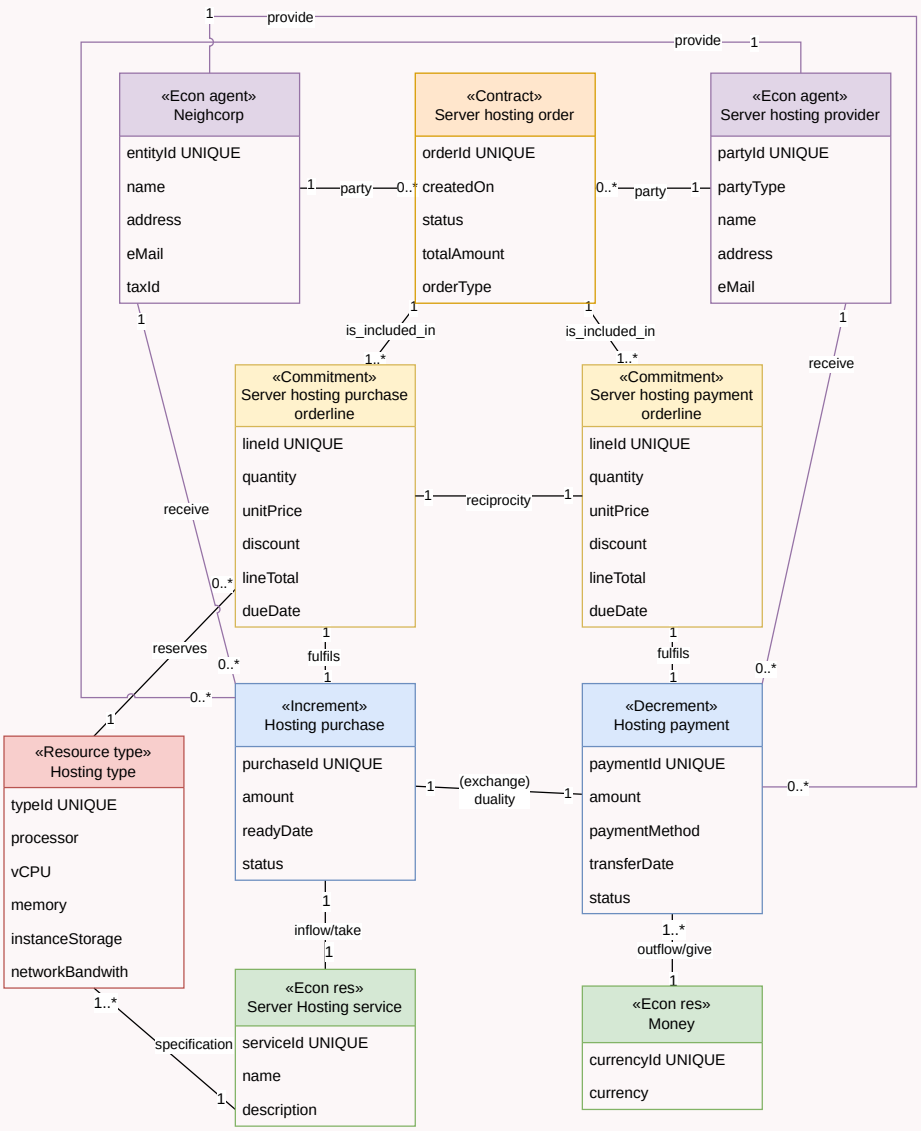


Figure 15: REA Server Hosting Exchange

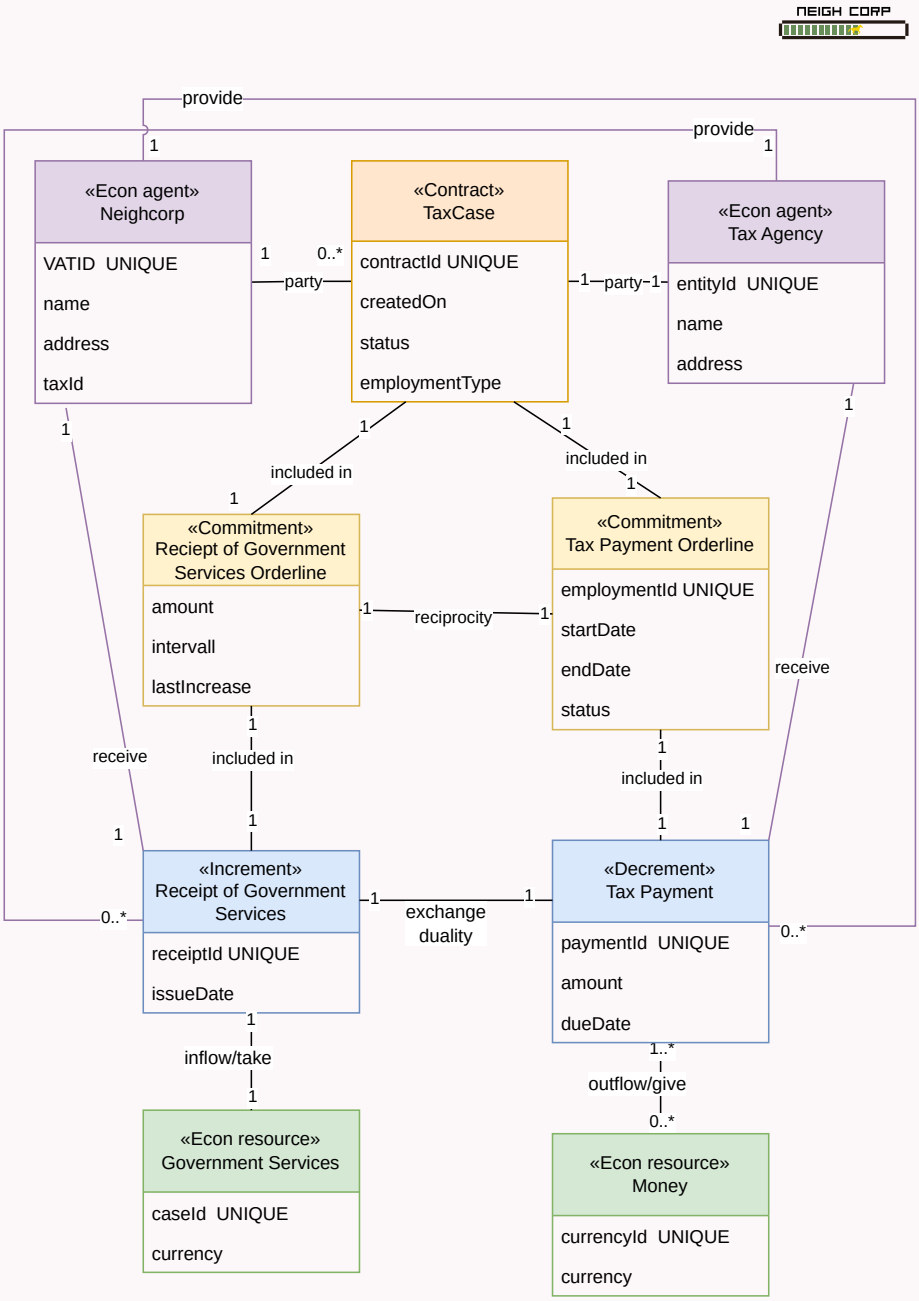


Figure 16: REA Tax Exchange

The Exchanges are all modelled in a similar way, with the main difference being added types for some economic resources (Hosting Type and Labour Type). We

describe the Premium Pass Exchange as an example.

The Premium Pass Exchange describes how Neigh Corp provides premium game access to a player in return for money. Neigh Corp and the player are the economic agents, while Premium Game Access and Money are the resources exchanged.

The Premium Pass Order represents the agreement between the two parties. It records the order ID, creation date, status and total amount. The order includes two commitments: the Premium Pass Purchase Orderline and the Premium Pass Payment Orderline. These represent the promise to provide premium access and the promise to pay for it. The reciprocity relationship connects these commitments because each is made in return for the other.

The commitments are fulfilled through two economic events. The Premium Pass Assignment records the actual allocation of premium access to the player's account, including the account ID, activation date and status. The Premium Pass Payment records the payment received, including the payment ID, amount, payment method, transfer date and status.

From Neigh Corp's perspective, the assignment is a decrement event because access is provided to the player, while the payment is an increment event because money is received. The exchange duality relationship connects these events as the two sides of the exchange. The model therefore distinguishes between what the parties promise and what they actually provide.

4.2 Detailed Processes in EPC

4.2.1 Sales Process

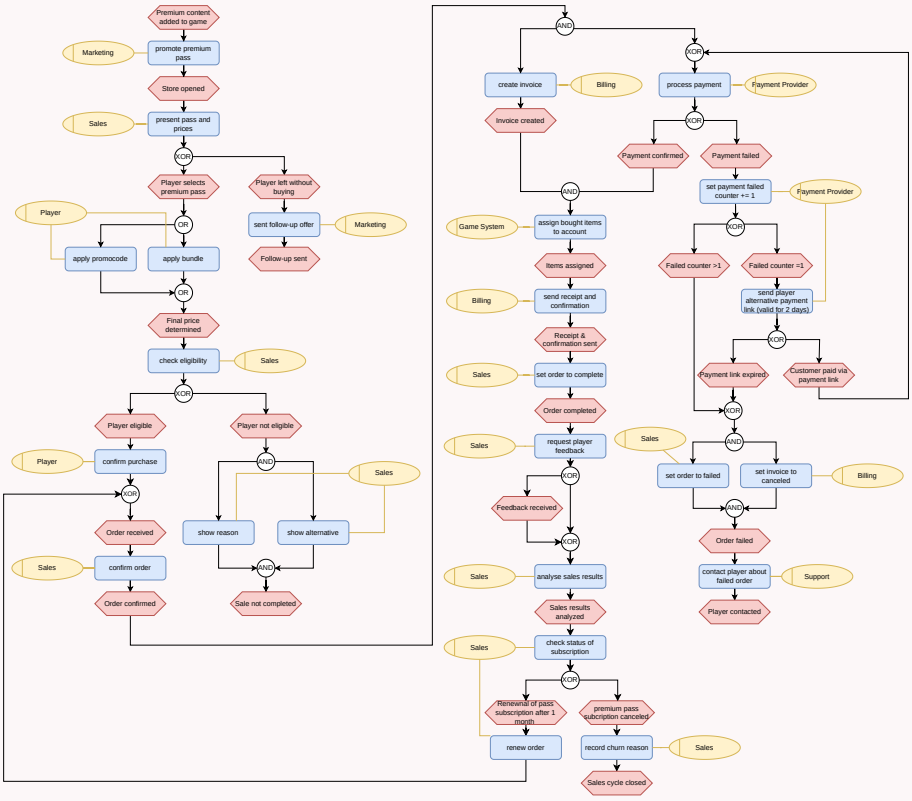


Figure 17: Sales Pocess *Neigh Corp*

Involved in the Sales Process are the organisational units Marketing, Sales, Player, Billing, Payment Provider, Shop System, and the ERP System. The sales process describes how Neigh Corp sells premium passes to players, from promoting the pass to completing the order and handling subscription renewal.

The process starts when new premium content is added to the game. Marketing promotes the premium pass, and Sales presents the pass and its prices when the store is opened. The player can either select a premium pass or leave without buying. If the player leaves, Marketing sends a follow-up offer. If the player selects a pass, they can apply a promotional code, a bundle offer, or continue without either option. The final price is then determined.

Sales checks whether the player is eligible to purchase the pass. If the player

is not eligible, the Shop System shows both the reason and an alternative, and the sale is not completed. If the player is eligible, they confirm the purchase, and Sales confirms the order.

After the order is confirmed, Billing creates an invoice while the Payment Provider processes the payment. Both the invoice and a confirmed payment are required before the Game System assigns the purchased items to the player's account. Billing then sends a receipt and confirmation, and Sales marks the order as completed.

If the payment fails, the Payment System increases a failure counter. After the first failed attempt, the player receives an alternative payment link that is valid for two days. If the player pays through this link, the process returns to payment processing. If the link expires or the payment fails more than once, the order is marked as failed and the invoice is cancelled. Support then contacts the player about the failed order.

For completed orders, Sales requests player feedback and analyses the sales results, whether or not feedback is received. The ERP System then checks the subscription status. If the subscription is renewed after one month, a new order enters the order-handling process. If the subscription is cancelled, Sales records the reason for cancellation and the sales cycle ends.

4.2.2 Procurement Process

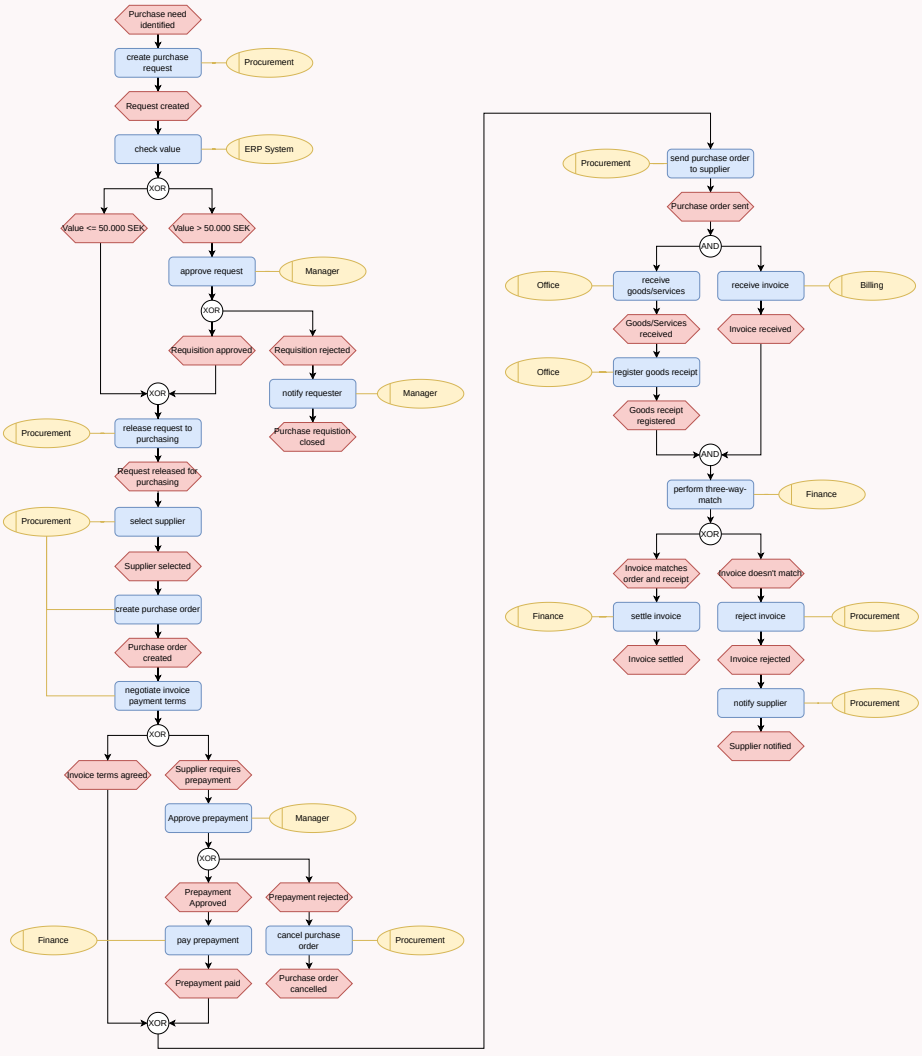


Figure 18: Procurement Processes *Neigh Corp*

In the Procurement Process, the organisational units Procurement, ERP System, Manager, Office, and Finance are involved.

The process starts when the company identifies a need to purchase goods or services. The Procurement department creates an internal purchase request, which is then registered. The ERP system checks the value of the request. If the value

exceeds SEK 50,000, a manager reviews the request and either approves or rejects it. Requests of SEK 50,000 or less continue without managerial approval. Approved requests and requests that do not require approval are released for purchasing. The Procurement department then selects a supplier. If the manager rejects the request, the ERP system notifies the requester and the case is closed.

After the supplier is selected, the Procurement department creates a purchase order and negotiates the payment terms with the supplier. If the terms are agreed without prepayment, the order is sent to the supplier. If the supplier requires prepayment, a manager must approve it. If approved, the Finance department pays the prepayment, and Procurement then sends the purchase order to the supplier. If the manager rejects the prepayment, Procurement cancels the purchase order and the process ends.

After the purchase order is sent, the Office receives the goods or services and registers the goods receipt. In parallel, the ERP system receives the invoice. Both steps must be completed before the process continues. The Finance department then compares the purchase order, the goods receipt and the invoice through a three-way match. If they match, Finance settles the invoice. If they do not match, the Procurement department rejects the invoice and notifies the supplier.

4.2.3 Hiring Process

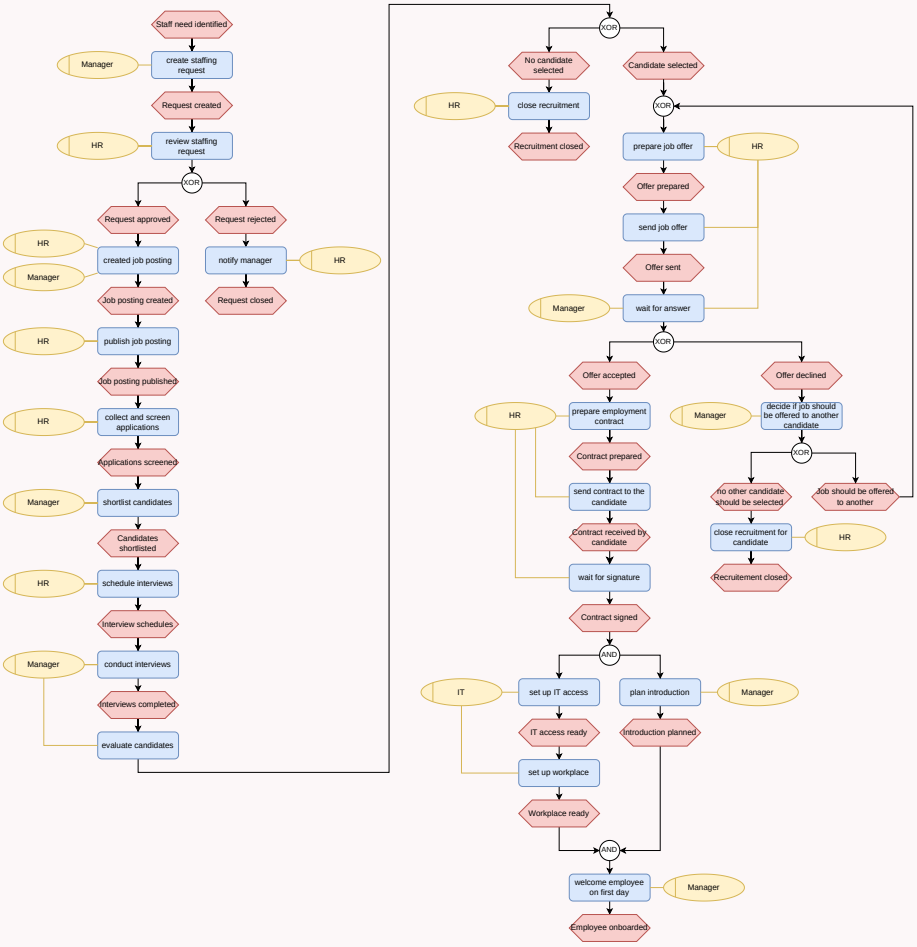


Figure 19: Hiring Process *Neigh Corp*

In the Hiring Process, the involved organisational units are HR, Manager, and IT.

When a need for staff is identified, the Hiring Process begins. A manager creates a staffing request which is being reviewed by HR. If they do not approve, the manager is notified and the request will be closed. If they approve, they create a job posting with the manager that they publish. Then, the applications will be collected and screened by HR. The manager shortlists the candidates and HR schedules interviews. The manager conducts interviews and evaluates

the candidates. If there are no fitting candidates to select, HR closes the recruitment. If one candidate get selected, HR prepares a job offer, sends it to the candidate and HR and the manager wait for an answer. If the candidate declines the offer, the manager decides whether to close the recruitment if no other candidate should be selected, or if the job should be offered and sent out to another candidate.

When an offer is accepted by a candidate, HR prepares and sends out an employment contract. After the contract is signed, IT sets up IT access and sets up the workspace while the manager plans an introduction for the new employee. Then, the new employee is welcomed on their first day by the manager. The process is completed now that the employee is onboarded.

4.2.4 Beta Testing Process

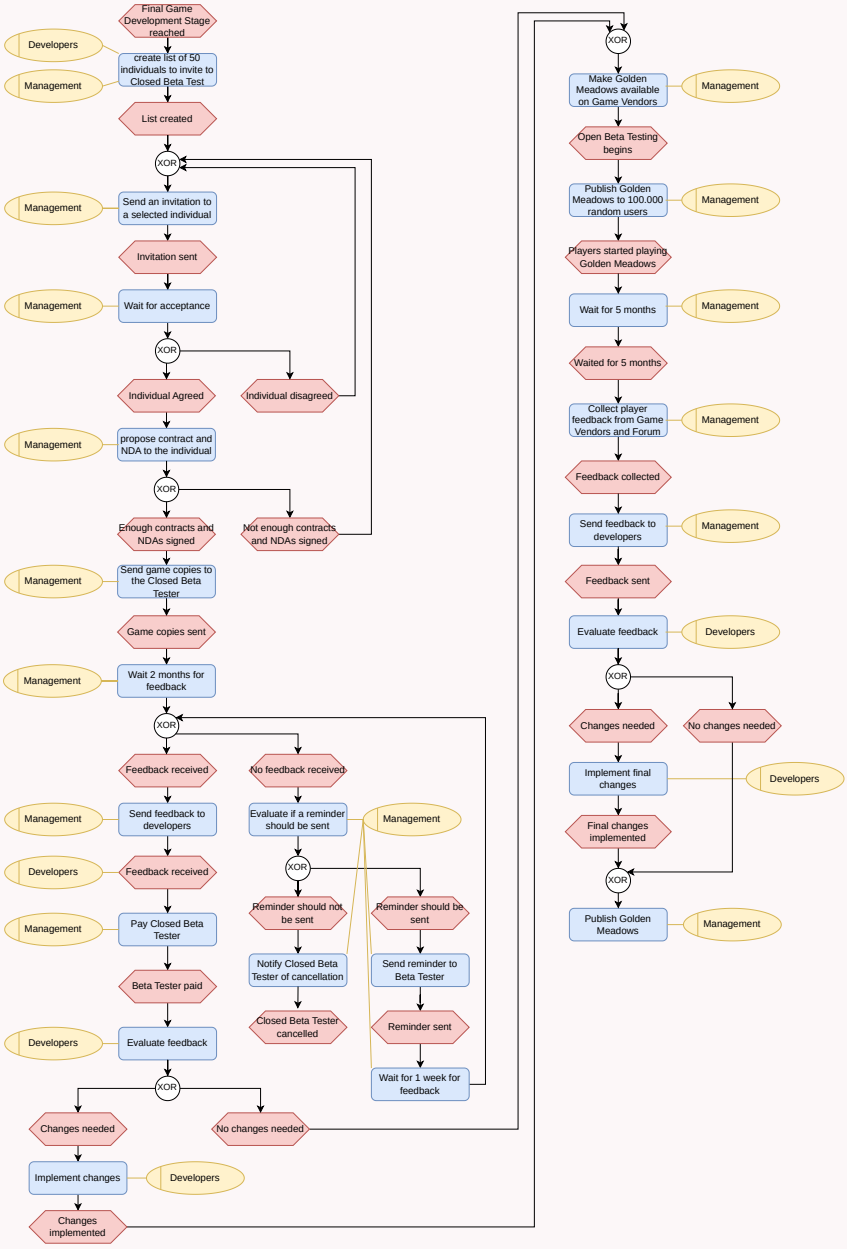


Figure 20: Beta Testing Process *Neigh Corp*

In the Beta Testing Process consists of the Sub-Processes Closed Beta Testing and Open Beta Testing. When the final Game Development Stage is reached, the Closed Beta Testing begins. The management and developer team selects a group of 50 individuals that are supposed to play Golden Meadows for the first time. Individuals from the list are selected and receive invitations from the Management, that they agree or disagree to. If they disagree, another individual is asked. If the individual agrees, the Management proposes a contract and a NDA to them. If they don't sign them, someone else from the list is asked. If they agree and sign, the Management sends a game copy to the Closed Beta Tester and waits for their feedback for two month, as agreed in the contract.

If feedback is received, the Closed Beta Tester is paid and the Management sends it to the developers. If no feedback is received after two months, the Management decides if a reminder should be sent to the Closted Beta Tester. If so, they will be notified and the Management waits another week. If no additional reminder should be sent, the Closed Beta Tester is simply notified of the cancellation of the process.

The developers evaluate the feedback and base new implementations and changes on them if needed. Now, the Closed Beta Testing is completed with making Golden Meadows available on the desired Game Vendors. The Open Beta Testing starts by publishing the game to 100.000 randomly selected Players who get early access to the game. After five months, all their feedback from the Game Vendors and the Game Forum is collected and sent to the developers by the Management. The developers evaluate the feedback, implement final changes if needed and finally Golden Meadows is published for all Players.

4.3 Relationships between Models

Include one or several tables that visualize the relationships between the models introduced in this section and the models in the previous sections. The tables should help to answer the following questions:

- How does the design of the processes help in fulfilling the goals of the company?
- What are the relationships between the exchange and conversion processes and the EPC models?
- What are the relationships between the EPC models and the VDML diagram?

Table 1: Traceability between EPC Models, REA Processes, and the VDML Diagram

EPC Process	**Related REA Exchange Processes**	**Related REA Conversion Processes**	**Corresponding VDML Actors & Value Propositions**
Sales Process	**Premium Pass Exchange:** Operationalizes the *Premium Pass Order* commitment, the *Premium Pass Payment* (increment of Money), and the *Premium Pass Assignment* (decrement of Premium Game Access).	**Brand Awareness Conversion:** Promotional activities and follow-up offers in the store utilize *Brand Awareness* and *High-Quality Game Content* to convert players into buyers.	**Actors:** Pay-to-Play Players, Payment Provider. **VPs:** Delivers *Premium Game Access VP* to players in exchange for *Money VP*; utilizes *Payment Service VP* from the Payment Provider.
Procurement Process	**Server Hosting, Insurance, Pentest, & Game Dev Exchanges:** Handles the purchase requests, PO commitments, 3-way matching, and invoice settlement (*Money* decrement) for external services and hardware.	**High-Quality Game Content & Merch Conversions:** Acquires the *Hardware*, *Software*, *Licenses*, and *Printed Material* resources consumed or used in internal conversion events.	**Actors:** Server Provider, Outsourced Dev Contractor, Penetration Testers, Insurance Company, Third-Party AI Vendor. **VPs:** Exchanges *Money VP* for *Server Hosting VP*, *Game Development VP*, *Pen-Test Results VP*, *Insurances VP*, and *AI Services VP*.
Hiring Process	**Labour Service Exchange:** Establishes the employment contract commitment through which employees provide *Labour Service* (increment) in exchange for *Money* (decrement).	**All Conversion Processes:** Supplies the *Labour Service* resource consumed across *Game Mechanics Development*, *Designing Textures/Animations*, and *Designing Motifs/Articles*.	**Actors:** Employees. **VPs:** Formalizes the exchange where Neigh Corp receives *Labour VP* from Employees in return for *Money VP*.
Beta Testing Process	**Base Game Access Exchange:** Operationalizes providing early game copies/access (decrement) in exchange for tester feedback (increment) and monetary compensation for closed testers.	**High-Quality Game Content Conversion:** Developer evaluation of beta feedback directly drives the *Game Mechanics Development* event to refine and produce *High-Quality Game Content*.	**Actors:** Beta Testers, Free-to-Play Players, Game Vendor, Game Forum. **VPs:** Exchanges *Early Access VP* and *Base Game Access VP* for *User Feedback VP* and *Anticipation VP*.

4.3.1 How Process Design Fulfills Company Goals

Table 2 demonstrates how the designed REA and EPC processes directly contribute to fulfilling Neigh Corp's strategic goals and means from the Strategy Map (Section 3).

Table 2: Contribution of Business Processes to Company Goals

Company Goal	**Supporting Process**	**Model Evidence**	**Contribution to the Goal**
Goal 1: To profit from Golden Meadows **Goal 8:** To generate recurring revenue **Goal 4:** To get users to make payments inside of Golden Meadows	• Sales Process (EPC) • Premium Pass Exchange (REA) • Procurement Process (EPC)	**Sales EPC:** Includes automated ERP monthly subscription status checks, promotional/bundle offers, and a 2-day payment retry link. **Procurement EPC:** Enforces a SEK 50,000 managerial approval threshold and 3-way match. **Goal Model:** Implements Means 13, 15, and 17.	Directly generates recurring revenue by automating monthly subscription renewals and recovering failed transactions via alternative payment links, while the Procurement process controls expenditure to protect profit margins.
Goal 2: To make Golden Meadows enjoyable to play for the target market **Goal 7:** To ensure the same quality of game mechanics for all	• Beta Testing Process (EPC) • High-Quality Game Content Conversion (REA) • Character Design Conversion (REA)	**Beta Testing EPC:** Two-stage feedback loop (50 Closed Beta Testers under NDA + 100,000 Open Beta Players) evaluated by developers before release. **REA Conversions:** Consumes *Labour Service* and *Hardware* to create game mechanics, textures, and animations. **Goal Model:** Implements Means 1, 27–33, and 36.	Ensures bugs and balancing issues are resolved prior to full launch and that core equestrian mechanics (breeding, genetics, competitions) and visuals meet target audience expectations.

Company Goal	Supporting Process	Model Evidence	Contribution to the Goal
Goal 3: To gain a loyal community **Goal 5:** To get many horse-interested users to play **Goal 9:** Establish a strongly recognizable brand	• Brand Awareness Conversion (REA) • Merch Design & Merch Conversion (REA) • Base Game Access Exchange (REA)	**REA Conversions:** Combines *Selling Merch*, *Publishing Game Content*, and *Publishing Social Media Content* into *Awareness Acquisition*. **Base Game Exchange:** Provides free access in exchange for user feedback. **Goal Model:** Implements Means 2, 3–9, and 26.	Lowers the switching cost to zero via free-to-play access while building a recognizable niche identity through merchandise, social media presence, and early community involvement on Steam and the Game Forum.
Goal 10: To have high long-term user retention **Goal 6:** To build a large player base	• Sales Process (EPC) • Hiring Process (EPC) • Game Development Exchange (REA)	**Sales EPC:** Captures post-purchase player feedback and logs cancellation reasons when subscriptions are not renewed. **Hiring EPC & Game Dev Exchange:** Onboards internal staff and external contractors to maintain a steady pipeline of content.	Analyzing cancellation reasons and player feedback allows Neigh Corp to address churn drivers, while continuous staffing and outsourced development ensure regular content updates that keep players engaged long-term.

5 IT Architecture Design

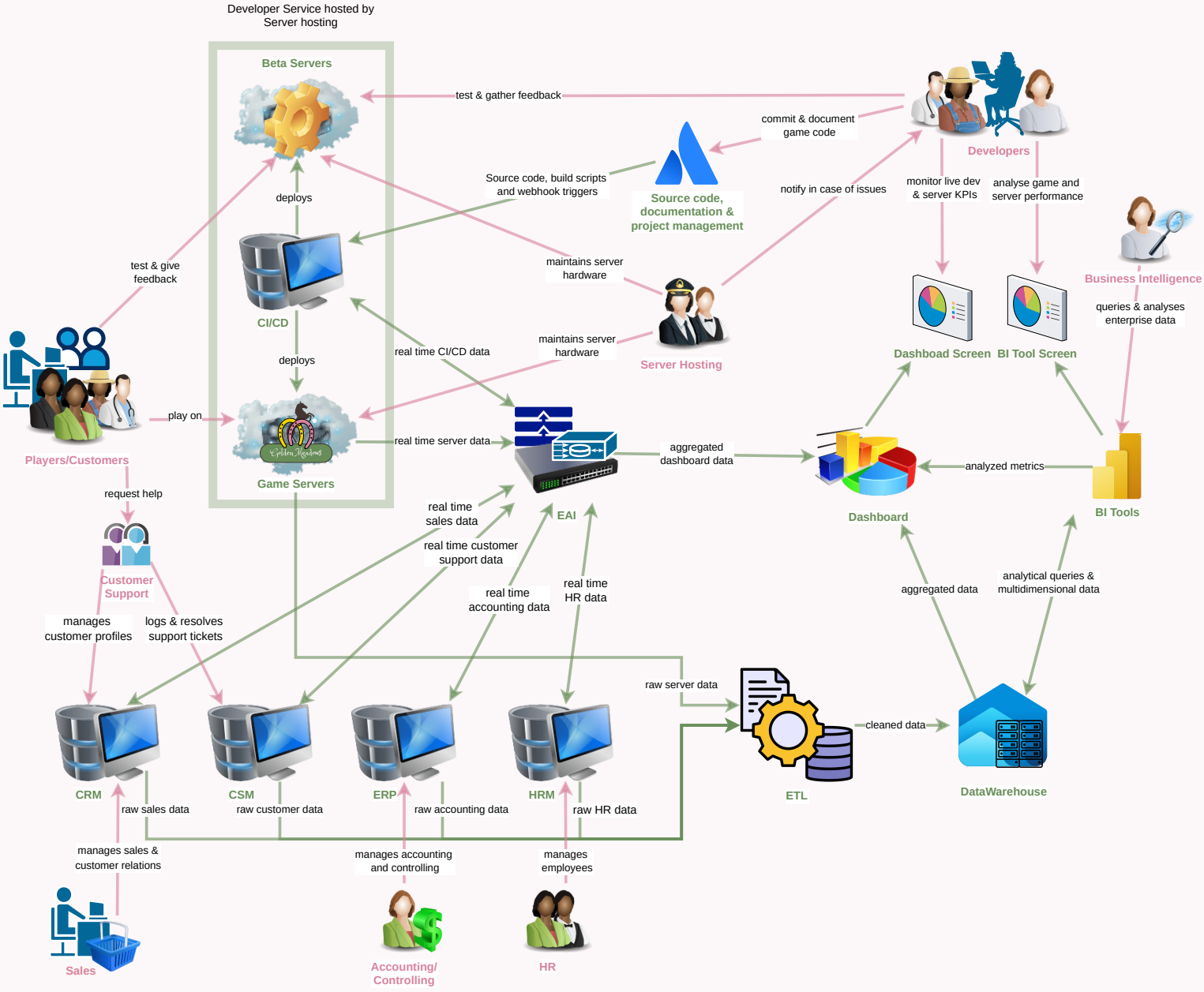


Figure 21: IT Architecture Diagram *Neigh Corp*

Figure 21 shows the IT-architecture of Neigh Corp. Systems are represented as nodes and the relationships between them as arrows, with some arrows being bi-directional. The graph is generally divided into three sections: one for operational systems, one for development and integration and an analytical section.

At the bottom of the graph there are the two systems: ETL and DataWarehouse. These are responsible for receiving raw data from the operational systems, ranging from CSM to HRM, aggregating these and storing them for future evaluation. The data warehouse stores current and historical data and aggregates these for the Dashboard node and BI Tools. In this case, the data warehouse is used for longer running measurements and KPIs, such as income within the last 12 months and whether break even is reached by year two, seen in Table 3 below. On the other hand, the EAI system handles real time information and shorter term KPIs. The EAI outputs its data to real-time dashboards that developers monitor continuously. This real time updating lets them react quickly to outages and bugs and therefore minimize monetary loss.

The systems feeding the ETL and EAI with data are the operational systems of the company, CSM, ERP, CRM and HRM. These are each interacted with by employees to fulfil their function. Among others, a system such as the CRM system is used for sales by the Sales department. This supports the process of selling the customer a premium pass, displayed in Figure 17 above. The CRM system is also connected to the Customer Support team so that these can use the customer database to provide assistance. In the operational layer there are also systems such as HRM which is used by HR to manage employees which, among others, supports the hiring process specified in Figure 19.

The hosting of the game is split between two servers: the normal game servers for everyday players and the Beta Servers where new versions of the game are tested. These are used by players as the graph shows. These two servers are connected to a CI/CD pipeline that enables continuous deployment, and all three of these nodes are hosted on the Server providers hardware. T

6 Business Performance Management

This section is to introduce the key performance indicators (KPI) for measuring and analysing the business processes of the company.

6.1 Leading and Lagging KPIs

Table 3: KPI definitions

Name	**Definition**	**Goals related to**	**Target values**	**Measured business process**	**IT system for data source**	**IT solution feeding the dashboard**
Time of beta testing completion	Lagging KPI	Goal 7, Objective 7	$\geq$ 1 month before release	Beta Testing Process Model	Game Server, Source code & documentation & project management	EAI
Golden Meadows income within last 12 months	Lagging KPI	Goal 1, Objective 6	$\geq$ 1 000 000 000 USD income	—	ERP, CRM	Data Warehouse
Golden Meadows income break even	Lagging KPI	Goal 1, Objective 6	Break even by end of year two	—	ERP, CRM	Data Warehouse
Time of attaining specific microtransaction count	Lagging KPI	Goal 4, Objective 3	$\geq$ 4000 microtransactions, $\leq$ year 1	—	ERP, CRM	EAI
Follower count on Steam pre-release	Lagging KPI	Goal 3, Objective 1	$\geq$ 9000 followers	—	ERP	EAI
Amount of monthly subscriptions	Lagging KPI	Goal 8	$\geq$ 200 000	—	CRM, ERP	EAI
Revenue generated by microtransactions monthly	Lagging KPI	Goal 1, Goal 4	$\geq$ 83 Million USD	—	CRM, ERP	EAI
Maintain retention rate monthly	Leading KPI: influences revenue	Goal 1, Goal 4	$\geq$ 90 %	—	CRM	EAI
Number of scheduled beta tests	Leading KPI: beta testing completion time	Goal 7, Objective 7	$\geq$ 100 test sessions	Beta Testing Process	Source code & documentation & project management	EAI
Change in Steam rating within last month	Leading KPI: Steam follower count	Goal 2, Objective 8	$>$ mediocre average rating	—	ERP	EAI

Continued on next page

Name	Definition	Goals related to	Target values	Measured business process	IT system for data source	IT solution feeding the dash-board
Very high Steam rating within a year	Lagging KPI	Objective 8, Goal 2	Highest rating	—	ERP	EAI

6.2 Description of KPIs

We defined 8 Lagging and 3 Leading KPIs for monitoring our progress. All stand in relationship with the EAI that feeds the dashboard. Each KPI is related to one or more company goals, specified with target values and mapped to the IT system.

- **Golden Meadows income break even** - Lagging KPI related to Goal 1 (To profit from Golden Meadows) as well as Objective 6 (One billion dollars annual income 2 years after the release, or reach break even within 2 years). The only identified target value is to break even by end of year two, the related IT solution being EAI and the data sources identified being ERP CRM. The measured business process is Beta Testing Process Model.
- **Time of attaining specific microtransaction count** - Lagging KPI concerning Goal 4 and Objective 3 (To get users to make payments inside of Golden Meadows and to receive 4000 daily microtransactions by the end of year 1). The target value is derived from the objective and is simply to attain more than or exactly 4000 microtransactions no longer than one year after release. The same IT data sources are used as in the previous KPI, as well as the same IT solution.
- **Follower count on Steam pre-release**, also identified as a lagging KPI, related to Goal 3 (To gain a community that is interested in all of the company's games) and Objective 1 (Have 9.000 followers on Steam before the release). Target value: more than or exactly 9000 followers. ERP is the only data source while the IT solution is the same as in the previous two lagging KPI's.
- **Amount of monthly subscriptions** - Lagging KPI related only to goal 8 (To generate recurring revenue from the player), sharing the same data sources and the IT solution as the first two identified KPI's. Target value is 200000, i.e. the amount of subscribers.
- **Revenue generated by microtransactions monthly** - related goals 1 4 (To profit from Golden Meadows and To get users to make payments inside of Golden Meadows), also being lagging, having the target value of

no less than 83 million USD. The same data sources and the IT solution are used as before.

- **Maintain retention rate monthly** - Leading KPI related to the same goals as the previous KPI, only using CRM as its data source while also using EAI as the IT solution. Its target value is more than or equals to 90%, representing retention rate. This KPI influences lagging KPI's related to revenue.
- **Number of scheduled beta tests** - Leading KPI influencing *Time of Beta Testing Completion*, data sources being source code, documentation, as well as project mangement. The target value is no more than 100 test sessions. The measured business process is Beta Testing Process while the IT solution is EAI. It relates to goal 7 (to ensure the same quality of game mechanics for all) and objective 7 (complete beta testing 1 month before release).
- **Change in Steam rating within last month** - Third leading KPI influencing *Steam Follower Count*. Its target value is greater than mediocre/average rating, data source being ERP and IT solution being EAI. It relates to goal 2 (to make Golden Meadows enjoyable to play for the target market) as well as objective 8 (reach 4 out of 5 stars for Golden Meadows on Google Play 3 months after the release AND reach the "very positive" rating on steam within one year after release).
- **Very high Steam rating within a year** - Lagging KPI related to the same goal and objective as the previous one, its target value being "highest rating", data source and IT solutions also being identical to the ones used by the leading KPI above.

6.3 Evaluation of KPIs

The provided KPIs in the previous section are aligned with the evaluation Eckerson proposes in "Performance Dashboards: Measuring, Monitoring, and Managing Your Business".

The twelve evaluation points that are listed are aligned, owned, predictive, actionable, few, easy, balanced & linked, trigger changes, standardized, context-driven, reinforced with incentives, and relevant.

The KPIs are designed to be aligned with the company's goals and objectives, with each KPI linked to specific business outcomes. They include both leading and lagging indicators, allowing the company to monitor factors that influence future performance as well as measure achieved results. The target values provide measurable performance expectations, while the identified business processes and IT systems indicate where the relevant data can be obtained and how it can be integrated into the dashboard.

Furthermore, the KPIs cover several areas of business performance, including

financial results, customer retention, marketing performance, and beta testing. This supports a balanced view of the company's progress. The defined targets can also help identify deviations from expected performance and indicate when corrective action may be necessary.

However, not all evaluation points can be confirmed from the KPI definitions alone. Criteria such as clear ownership, ease of understanding, standardized measurement, contextual interpretation, and the use of incentives require additional information about how the KPIs will be implemented and managed. Therefore, while the proposed KPIs address several of Eckerson's evaluation principles, their full compliance with all twelve criteria depends on the company's broader performance management practices.

7 Enterprise Architecture Evaluation

In other parts of the work, you have designed the organization from several perspectives: goals, processes, IT architecture, and so on. All of these combined converge towards an Enterprise Architecture (EA). A key component of EA is capability, an ability and capacity that enable an enterprise to achieve a business goal in a certain context. In this light, your task is to design one capability based on the models already created. The first step is to fill in the capability elicitation template in terms of relevant goals, KPIs, context, business process variants. This should be followed by a graphical design of a capability model, for instance, by introducing links between the newly developed capability and the corresponding constructs in other models.

Table 4: Capability: Game Development

Field	**Description**
Capability Name	Game Development
Capability Description	The ability to plan, design, develop, test, and deliver high-quality game content that makes Golden Meadows enjoyable, appeals to horse-interested users, maintains consistent game mechanics, and encourages players to continue playing.
Relevant Business Goals	• Goal 1: To profit from Golden Meadows. • Goal 2: To make Golden Meadows enjoyable to play for the target market. • Goal 7: To ensure the same quality of game mechanics for all. • Goal 10: To have high long-term user retention.



Field	Description
Key Performance Indicators (KPIs)	• Time of beta testing completion • Maintain retention rate monthly • Number of scheduled beta tests • Change in Steam rating within last month • Very high Steam rating within a year
Context	Golden Meadows is a MMORPG Horse game developed by Neighcorp. The target market is horse interested persons that are young and want a realistic Horse experience. The company develops key functionalities itself, but also outsources parts of the Development. This can lead to variating quality.
Business Process Variants	• High Quality Game Content Conversion Process • Character Design Conversion Process • Labour Service Exchange Process • Pentest Service Exchange • Hiring Process • Procurement Process • Beta testing process
Related VDML Value Exchanges	• Beta testers - Neighcorp • Third Party AI vendor - Neigcorp • Employees - Neighcorp • Outsourced development contractor - Neighcorp • 3rd Party authorized cosmetics creators on marketplace - Neighcorp • Game Forum - Neighcorp
Supporting IT Architecture	Game Development is supported by databases for storing game-related data, servers for hosting and running game services, and ETL processes for transferring and transforming data. Business intelligence (BI) tools and the data warehouse can support analysis of game performance and player behaviour. Development tools and testing environments also support the creation, testing, and improvement of game content.

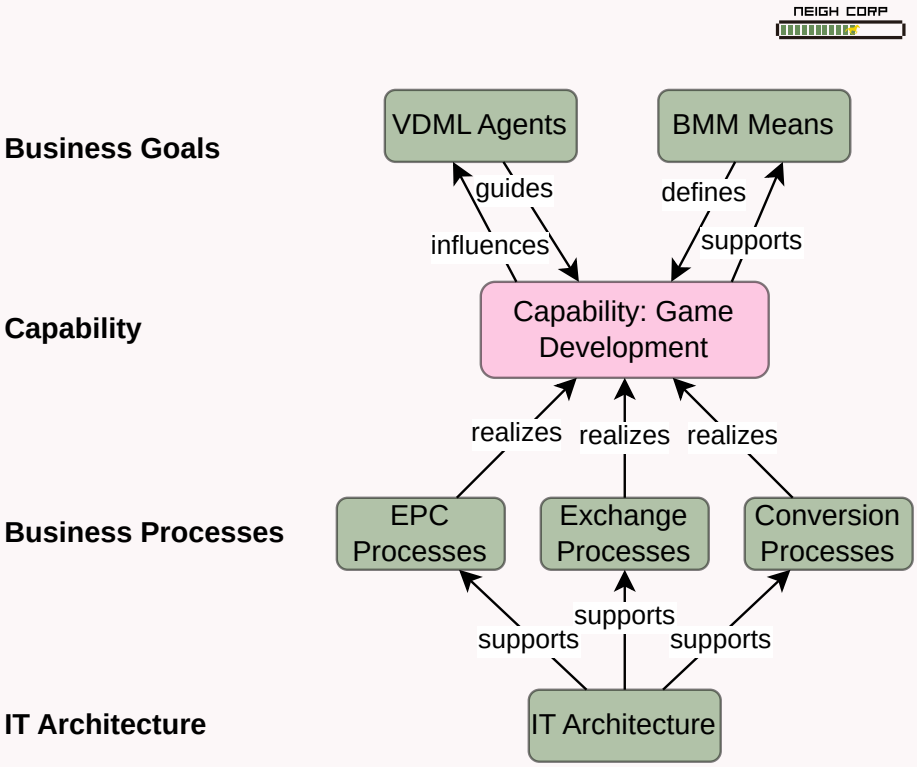


Figure 22: Capability Map

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[1] Robert S. Kaplan and David P. Norton. *Strategy Maps: Converting Intangible Assets Into Tangible Outcomes*. Boston, MA: Harvard Business School Press, 2004. ISBN: 978-1591391340.

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